



Regulation of Stem Cell Research

Argentina • Brazil • Chile • Costa Rica
El Salvador • Mexico • Panama

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Comparative Summary

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This multinational report covers stem cell research in the selected jurisdictions of **Argentina, Brazil, Chile, Costa Rica, El Salvador, Mexico, and Panama**. The report focuses on regulation of scientific research in general, regulation of stem cell research, and relevant legislative developments.

Brazil and **Mexico** have constitutional principles for the support of scientific and technological research. They have enacted legislation to implement support of such research, as have **Argentina** and **El Salvador**. **Chile** and **Panama** have legislation on scientific research, and **Costa Rica** has legislation regarding medical research.

All the selected jurisdictions include in their legislation the need to obtain informed consent from persons undergoing some sort of treatment by explaining the procedure and risks involved before they are submitted to it. **Argentina, Brazil, Chile, Costa Rica, and El Salvador** have provisions in their laws requiring that all institutions involved in scientific research have an ethics committee to evaluate and uphold medical standards.

Regarding the use of stem cell therapy, **Brazil** only allows the use of embryonic stem cells produced by in vitro fertilization when certain conditions are met. **Chile** does not allow the destruction of human embryos obtained from stem cells for the culture of tissues and organs. **Costa Rica** and **El Salvador** permit the therapeutic use of adult stem cells, while prohibiting the use of embryonic stem cells. **Panama** does not allow the creation of human embryos to obtain stem cells. Figure 1 depicts whether the use of embryonic stem cells is permissible in the surveyed jurisdictions under current law.

The Supreme Courts of **Brazil** and **Costa Rica** have been involved in disputes regarding the use of stem cells in therapeutic treatments. **Brazil's** Supreme Court has fully authorized the use of embryonic stem cells for therapeutic purposes, while **Costa Rica's** Supreme Court has upheld the ban on the use of stem cells, but authorized its use on people who were already receiving the stem cell treatment.

Relevant bills have been introduced in the legislative bodies of **Brazil, Chile, and Panama**. The bill in **Brazil** proposes a total ban on the use of stem cells obtained from human embryos for any purpose. The bill in **Chile** aims to promote scientific research in cell therapy and stem cells, and the bill in **Panama** proposes a new regulatory regime for stem cell-based treatments.

In **Mexico**, the Department of Health has published a proposal addressing several therapeutic matters, including a section on stem cell research for the purpose of, among other things, prohibiting the false advertising of cures for many types of diseases based on unproven cell-based treatments.

Figure 1: Use of Embryonic Stem Cells



Argentina

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SUMMARY Argentina has national and provincial legislation promoting the development of scientific research in general. A national law of 2001 protects initiatives on scientific and technological research, establishing priority plans and programs. The law sets forth principles, according to which both public governmental agencies and the private sector are to conduct scientific research. Several government agencies and commissions are established at the national level. The provinces also have their own scientific research institutions. Another national law of 1996 mandates the creation of hospital ethics committees at every public hospital. Another national law of 2013 creates the National Registry of Genetic Data for criminal purposes, but also addresses general principles concerning the gathering and use of genetic information. In addition, a decree promulgated in 1997 prohibits cloning experiments related to humans. Moreover, a national law of 2000 established a registry of hematopoietic progenitor cell donors, and its provisions are adopted and further developed at the provincial level, examples of which are cited in this report.

I. Regulation of Scientific Research in General

Argentina has passed legislation to protect human beings in light of the recent developments of scientific knowledge in the field, thus reacting to what is considered an excessive use of new technical means applied to the field of human health sciences.¹

A. Law No. 25,467 of 2001 on the National System of Science, Technology and Innovation

1. In General

Law No. 25,467 creates the National System of Science, Technology and Innovation, with the purpose of increasing the cultural, educational, social, and economic heritage of Argentinean society.² The main purpose of this initiative is to promote and guide scientific and technological research, establishing priority plans and programs,³ and to promote scientific and technological cooperative actions at the international level, with a special emphasis on the Mercosur region.⁴

¹ Jorge Nicolás Lafferriere, *Leyes sobre Bioética en Argentina*, Centro de Bioética Persona & Familia (Feb. 2, 2015), <https://perma.cc/L3HC-2ZK4>.

² Ley No. 25.467 de 29 de agosto de 2001, Sistema Nacional de Ciencia, Tecnología e Innovación, art. 1, <https://perma.cc/XC52-CH2Y>.

³ Id. art. 2(f).

⁴ Id. art. 2(i).

2. *Principles*

The following are among the most important principles for any research activity in science, technology and innovation: (a) respect for the dignity of the human person;⁵ (b) respect for the privacy and intimacy of research subjects and the confidentiality of the data obtained;⁶ (c) the free and voluntary participation of people in research trials;⁷ (d) the obligation to use informed consent processes prior to recruiting research subjects;⁸ (e) the obligation to carry out preclinical and animal trials prior to human experimentation;⁹ and (f) the prohibition on the commercialization of the human body, its parts, or genetic information of any kind.¹⁰

3. *Responsibilities of the State*

Per Law No. 25,467, the Argentinean state has the following responsibilities in matters of scientific, technological and innovation policy: (a) to create conditions for the generation of scientific knowledge;¹¹ (b) to provide guidance for scientific research and technological development, establishing priorities in strategic areas;¹² and (c) to establish the National Science, Technology and Innovation Plan, its priorities, and programs.¹³

4. *Establishment of Governmental Entities in the Area of Science, Technology and Innovation*

The law has created the following entities to promote scientific research in the country:

a. National System of Science, Technology and Innovation

The System is composed of the political organs providing guidance in these areas, as well as universities, other organizations, and entities at the national, provincial, and municipal levels, and also the private sector, which carry out substantive activities linked to scientific and technological development.¹⁴

⁵ Id. art. 3(a).

⁶ Id. art. 3(b).

⁷ Id. art. 3(c).

⁸ Id. art. 3(d).

⁹ Id. art. 3(e).

¹⁰ Id. art. 3(j).

¹¹ Id. art. 5(a).

¹² Id. art. 5(c).

¹³ Id. art. 5(e).

¹⁴ Id. art. 4.

b. Scientific and Technological Cabinet

The law also created the Scientific and Technological Cabinet, which is chaired by the chief of the Cabinet of Ministers and is composed of all the state secretaries with competency in science and technology.¹⁵ The Cabinet's functions are, in particular, to: (a) establish the National Science, Technology and Innovation Plan;¹⁶ (b) propose the annual budget for science and technology functions in the national appropriations budget;¹⁷ and (c) evaluate annually the execution of the National Science, Technology and Innovation Plan.¹⁸

c. Federal Council of Science, Technology and Innovation

The Council is made up of the highest-level officials in the areas of the provincial governments and the Autonomous City of Buenos Aires. The Council is tasked with the development and articulation of national and regional policies in the areas of scientific, technological, and innovative activities throughout the country.¹⁹

d. National Agency for Scientific, Technological and Innovation Promotion

The Agency has been created as a decentralized body dependent on the Secretariat for Technology, Science and Productive Innovation to develop instruments for the promotion and financing of scientific, technological, and innovative development projects in the country.²⁰

e. Interinstitutional Council of Science and Technology

The Council is composed of the highest authorities of the national organizations that carry out scientific and technological activities,²¹ with the following functions: (a) to contribute to compliance with best practices within the National Science, Technology and Innovation Plan;²² and (b) to improve links between the public and private sectors, promoting private sector participation in investments in science and technology.²³

¹⁵ Id. art. 8.

¹⁶ Id. art. 8(a).

¹⁷ Id. art. 8(b).

¹⁸ Id. art. 8(c).

¹⁹ Id. art. 10, para. 1.

²⁰ Id. art. 12.

²¹ Id. art. 14(a).

²² Id. art. 15(a).

²³ Id. art. 15(d).

f. Advisory Commission for the National Science, Technology and Innovation Plan

This is another government entity created to assist in the development of scientific research in Argentina.²⁴

B. Law No. 24,742 of 1996 on the Public Health System

This law mandates that every hospital of the public health and social security system must have a hospital ethics committee, with the functions of advising, studying, instructing, and supervising scientific research concerning ethical issues arising from the practice of hospital medicine.²⁵

These committees function as interdisciplinary teams made up of doctors, paramedical personnel, lawyers, philosophers, and professionals in the sciences of human behavior.²⁶ The committees' functions center on the following areas: (a) reproductive technologies, (b) eugenics, (c) human experimentation, (d) artificial prolongation of life, (e) euthanasia, (i) genetics, and (j) organ transplants.²⁷

Committees' recommendations are not binding, and do not exempt intervening professionals or hospital authorities from their ethical and legal responsibilities.²⁸

C. Law No. 26.879 on the National Registry of Genetic Data

This law creates the National Registry of Genetic Data, which is linked to crimes against sexual integrity, and operates within the scope of the Ministry of Justice and Human Rights.²⁹ The law forbids the use of deoxyribonucleic acid (DNA) samples for any purpose other than the identification of persons for the purposes provided for in the law.³⁰

The Registry's exclusive purpose is to facilitate the clarification of facts that are the subject of a judicial investigation in criminal matters linked to crimes against sexual integrity, in order to find the responsible persons.³¹

²⁴ Id. art. 16.

²⁵ Ley No. 24.742 del 27 de noviembre de 1996, Sistema Público de Salud, art. 1, <https://perma.cc/9QKR-WMKN>.

²⁶ Id. art. 2.

²⁷ Id. art. 3(a)-(e), (i) & (j).

²⁸ Id. art. 4.

²⁹ Ley No. 26.879 de 3 de julio de 2013, Créase el Registro Nacional de Datos Genéticos, art. 1, <https://perma.cc/VK95-5PEL>.

³⁰ Id. art. 11.

³¹ Id. art. 2.

The Registry stores and systematizes the genetic information associated with a sample or biological evidence that has been obtained in the course of a criminal investigation of any person convicted with a final sentence for specified crimes.³²

The registered genetic information consists of the personal alphanumeric record prepared exclusively on the basis of genotypes that segregate independently, are polymorphic in the population, that lack direct association in the expression of genes, and provide only identifying information suitable for being systematized, and are coded in a database.³³

The records in the Registry are considered sensitive and confidential data, and may be provided exclusively to the Public Prosecutor's Office, judges, and courts throughout the country in the context of judicial cases for specified crimes.³⁴

Genetic tests are carried out in laboratories accredited by the Ministry of Science, Technology and Productive Innovation or by certifying organizations duly recognized by that Ministry.³⁵ The Registry must ensure the inviolable and unalterable preservation of genetic information files and samples obtained.³⁶

The information in the Registry is to be deleted after one hundred years from the initiation of the case in which its incorporation was ordered or by court order.³⁷

II. Regulation of Stem Cell Research (*células madre*)

A. National Legislation

National legislation prohibits cloning experiments related to humans.³⁸

Law No. 25,392 of 2000 created the National Registry of Hematopoietic Progenitor Cell Donors,³⁹ which is based at the National Central Single Ablation and Implant Coordinating Institute (Institute).⁴⁰ The Registry serves as the repository of the identification and affiliation data of

³² Id. art. 3, para. 1.

³³ Id. art. 4.

³⁴ Id. art. 7.

³⁵ Id. art. 8.

³⁶ Id. art. 9.

³⁷ Id. art. 10.

³⁸ Decreto No. 200 de 8 de marzo de 1997, Prohíbense los Experimentos de Clonación Relacionados con Seres Humanos, art. 1, <https://perma.cc/4PSR-JCYC>.

³⁹ Ley No. 25.392 de 30 Noviembre de 2000, Registro Nacional de Donantes de Células Progenitoras Hematopoyéticas, art. 1, <https://perma.cc/U9BQ-3522>.

⁴⁰ Id. art. 2.

potential donors, and must also record all information derived from histocompatibility studies of hematopoietic progenitor cells carried out in accredited laboratories.⁴¹

The Institute is the enforcement authority of the National Registry of Hematopoietic Progenitor Cells and must develop education and dissemination programs for raising awareness among the population about the objective, principles, and subject matter of Law No. 25,392, in particular concerning: (a) donor recruitment centers, (b) the description of donors, and (c) the registry's data center.⁴² The Institute may exchange information with all countries having records similar to those created by Law No. 25,392.⁴³

B. Provincial Legislation

1. Chaco Province

Provincial Law No. 6,196 of 2008 creates the Bank of Stem Cells not included in Law No. 25,392, with the aim of cryopreserving them and protecting the future of the person.⁴⁴

The Provincial Ministry of Public Health acts as the enforcement authority establishing quality standards on the process of collection, processing, and conducting studies for saving lives, the storage of stem cells for a certain period of time, and to carry out relevant institutional communication campaigns.⁴⁵

The Bank must inform the person requesting the collection, processing, study, or storage of stem cells about: (a) the nature and health implications of the practice to be carried out and the state of progress of the science; and (b) the possibility of donating the deposited cells, for their public and universal purpose.⁴⁶

Stem cell units may only be collected after the concerned person signs the informed consent form established in the law, and under the control and responsibility of the respective provincial health authorities.⁴⁷

⁴¹ Id. art. 3.

⁴² Decreto Reglamentario No. 267, sobre el Registro Nacional de Donantes de Células creados por Ley No. 25.392 de 13 de febrero de 2003, Annex 1, art. 2, <https://perma.cc/JF8Q-342Q>.

⁴³ Law No. 25,392, art. 4.

⁴⁴ Ley No. 6.196 de 2008 de la Provincia de Chaco, art. 1, <https://perma.cc/23FR-LXSW>.

⁴⁵ Id. art. 2.

⁴⁶ Id. art. 3.

⁴⁷ Id. art. 4.

2. *Chubut Province*

The Chubut Province adheres to National Law No. 25,392 through provincial Law No. 5,228 of 2004.⁴⁸

3. *Río Negro Province*

The Río Negro Province adheres to National Law No. 25,392 and its Regulatory Decree No. 267 through provincial Law No. 3,852 of 2004.⁴⁹

4. *Tierra del Fuego Province*

The Province of Tierra del Fuego, Antarctica and the South Atlantic Islands adheres to National Law No. 25,392 through provincial Law No. 649 of 2004.⁵⁰

Per this law, the Provincial Health Ministry is the responsible authority to: (a) develop education programs raising awareness among the population about the objective, principles, and subject matter of National Law 25,392; and (b) create Centers for the Collection and Donation of Hematopoietic Progenitor Cells.⁵¹

C. Human Genome

1. *National Legislation*

No legislation at the national level was found.

2. *Provincial Legislation*

a. Law No. 712 of 2001 of the Autonomous City of Buenos Aires

This law prohibits discrimination against any person on the basis of genetic information or genetic services.⁵² The law guarantees and protects the right to dignity, identity, and integrity of all people in relation to their genetic heritage.⁵³

⁴⁸ Ley No. 5.228 de 10 de octubre de 2004, art. 1, <https://perma.cc/9H24-62EA>.

⁴⁹ Ley No. 3.852 de 29 de julio de 2004, art. 1, <https://perma.cc/HR32-EEEP>.

⁵⁰ Ley No. 649 de 18 de noviembre de 2004, Adhesión de la Provincia a la Ley Nacional No. 25.392, sobre Registro Nacional de Donantes de Células Progenitoras Hematopoyéticas, art. 1, <https://perma.cc/Q589-CR8E>.

⁵¹ Id. art. 2.

⁵² Ley No. 712 de 29 de noviembre de 2001 de la Ciudad Autónoma de Buenos Aires, de Garantías del Patrimonio Genético Humano, art. 1, <https://perma.cc/EFY9-M3LH>.

⁵³ Id. art. 2.

The following information is excluded from the term “genetic information”: (a) information concerning the sex or age of the person; (b) information about the person’s blood, urine, or chemical tests, unless these tests are genetic tests; and (c) information on physical examinations, and any other relevant information that allows a determination of the person’s present state of health.⁵⁴

The law prohibits the dissemination or making public of genetic information of persons by any means, with the exception of cases authorized by the interested party or judicially.⁵⁵ The law does not prejudice the normal practice of health professionals in the treatment of patients to whom genetic studies of any kind are necessary.⁵⁶

However, the law allows payments for genetic studies carried out on a person pursuant to a medical order when coverage of the service is assumed by an insurance company, a social services agency, a prepaid health company, or an occupational risk insurer.⁵⁷

Any violation of the genetic discrimination or genetic disclosure clauses triggers compensation for damages, in accordance with the Civil Code.⁵⁸

Insurance companies, social services agencies, prepaid medicine companies, or occupational risk insurers are prohibited from: (a) requesting genetic analysis prior to insurance coverage or health services; (b) requesting, collecting, exchanging, or purchasing genetic information; or (c) delivering, under any concept or condition, genetic information to other insurance companies, social services, prepaid medicine companies, or occupational risk insurers, or to a person or company that collects, compiles, publishes, or disseminates insurance information, nor to an employer with respect to his employees.⁵⁹

Employers incur unlawful employment practices when they: (a) fail or refuse to hire, or discharge any person, or otherwise discriminate with respect to compensation, terms, conditions, or privileges of employment on the basis of genetic information referring to the concerned person or their family; (b) limit, segregate, or classify employees in a manner that would deprive or tend to deprive them of employment opportunities, or otherwise adversely affect their status as an employee, due to genetic information regarding them, or their family members; or (c) request, require, collect, or purchase genetic information regarding an individual or family member.⁶⁰

⁵⁴ Id. art. 3.

⁵⁵ Id. art. 5.

⁵⁶ Id. art. 4.

⁵⁷ Id. art. 6.

⁵⁸ Id. art. 7.

⁵⁹ Id. art. 8.

⁶⁰ Id. art. 9.

The handling of genetic information concerning an employee's medical reports is subject to the strictest rules of confidentiality.⁶¹

The following are exceptions to the aforementioned rules: (a) with respect to an employee who is the subject of the information, at his request; (b) when pursuant to a court order, provided that the employer gives the employee adequate notice to challenge the court order, unless the court order also imposes confidentiality requirements; and (c) regarding the enforcement authority when investigating compliance with the law.⁶²

Public organizations are authorized to use genetic information for exclusively statistical purposes, guaranteeing the anonymity of the information and pursuant to public policies.⁶³

The law adopts the Universal Declaration on the Human Genome and Human Rights of UNESCO of November 11, 1997, as guidance for the regulation and interpretation of Law No. 712 of 2001.⁶⁴

b. Law No. 5165 of 2005 of the Catamarca Province

This law guarantees and protects the right to dignity, identity, and integrity of all persons in relation to their genetic heritage.⁶⁵

The law contains the following prohibitions: (a) any discrimination based on a person's genome;⁶⁶ (b) carrying out worrying (*preocupacionales*) genetic studies, exams for social services, prepaid medicine companies, work risk insurers, or similar;⁶⁷ and (c) disseminating or making public the genetic information of persons by any means without their consent, except in cases authorized by a court.⁶⁸

c. Ley No. 8953 de la Provincia de Córdoba

This law establishes and guarantees the inviolability of the human genome.⁶⁹

It also prohibits the use, consultation, and dissemination of genetic studies of people for some of the following purposes: (a) for the preparation of reports aimed at obtaining jobs; (b) as part of

⁶¹ Id. art. 11.

⁶² Id. art. 12.

⁶³ Id. art. 13.

⁶⁴ Id. art. 14.

⁶⁵ Ley No. 5.165 de 1 de septiembre de 2005, Garantía y Resguardo del Derecho a la Dignidad, Identidad e Integración de Todas las Personas en Relación a su Patrimonio Genético, art. 1, <https://perma.cc/AE53-PTMM>.

⁶⁶ Id.

⁶⁷ Id. art. 2.

⁶⁸ Id. art. 3.

⁶⁹ Ley No. 8.953 de 19 de septiembre de 2001, Inviolabilidad del Genoma Humano – Establecimiento y Garantización en la Provincia--Prohibiciones y Sanciones, art. 1, <https://perma.cc/RGD9-KAJ9>.

the routine medical examinations performed on employees for the duration of their employment relationship; (c) as an element of assessment, admissibility requirement, or categorization by insurance companies, retirement and pension fund administrators, occupational risk administrators, public or private entities that provide social security and medical care services, mutuals, and cooperatives that provide or administer health services; (d) for the creation of genetic banks or registries not legally authorized; or (e) for any other action that, regardless of its object, lacks legitimate interest and is likely to generate discriminatory treatment.⁷⁰

Excluded are studies carried out pursuant to a judicial decision or when there is express consent of the interested person, for reasons of health or scientific research.⁷¹

The Provincial Health Ministry constitutes the enforcement authority, and administers the Provincial Registry of Clinical Laboratory Practices of Human Genetics, which registers all professionals within the territory of the province who are responsible for the operation of health centers or medical departments, both public and private, that carry out human genetic studies or therapeutic practices of any nature that involve the application of techniques for manipulating the human genetic heritage.⁷²

⁷⁰ Id.

⁷¹ Id. art. 3.

⁷² Id. art. 4.

Brazil

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SUMMARY Brazil's Constitution states that the government must offer incentives for scientific development, research, scientific and technological training, and innovation. A federal law enacted in 2004 establishes measures to encourage innovation and scientific and technological research.

The use of stem cells for research and therapy purposes is regulated by the federal Biosafety Law and a decree enacted in 2005. Different agencies have enacted regulations to further control the use of stem cells. In 2008, the Federal Supreme Court ruled on the constitutionality of the 2005 Biosafety Law, authorizing the scientific use of embryonic stem cells for therapeutic purposes. A bill of law introduced in the Federal Senate in 2020 seeks to amend the Biosafety Law and to prohibit, for any purpose, the use of stem cells obtained from embryos.

I. Regulation of Scientific Research in General

A. Constitutional Principle

Article 218 of Brazil's Constitution states that the nation must promote and give incentives to scientific development, research, scientific and technological training, and innovation.¹ Basic scientific research and technology must receive priority treatment from the state, considering public well-being and progress in science, technology, and innovation.² Technological research must be oriented principally toward solution of Brazilian problems and toward development of national and regional productive systems.³ The state must support human resources training in the areas of science, research, technology, and innovation, including by means of support for technological extension activities. The state must offer those engaged in such activities special means and conditions of work.⁴ The law must support and foster companies that invest in research, in creation of technology appropriate for the country, and in training and improvement of their human resources. Companies that adopt compensation systems that assure employees, apart from their salary, participation in the economic gains resulting from the productivity of their labor must also be supported by the law.⁵ The states and Federal District may allocate part of their budgetary receipts to public entities for promotion of education, as well as scientific and

¹ Constituição Federal art. 218, <https://perma.cc/D8KL-2TYJ>.

² Id. art. 218, § 1.

³ Id. art. 218, § 2.

⁴ Id. art. 218, § 3.

⁵ Id. art. 218, § 4.

technological research.⁶ In the execution of the activities provided for in article 218, the state must encourage coordination among entities, as much public as private, in the diverse spheres of government.⁷ The state must promote and give incentives for performance abroad by public institutions of science, technology, and innovation, with a view toward execution of the activities provided for in article 218.⁸

B. Technological Innovation Law

Law No. 10,973 of December 2, 2004, establishes measures to encourage innovation and scientific and technological research in the productive environment, with a view to technological training, the achievement of technological autonomy, and the development of the country's national and regional productive system.⁹

The measures must comply with the following principles:

I - promotion of scientific and technological activities as strategies for economic and social development;

II - promotion and continuity of scientific, technological and innovation development processes, ensuring human, economic and financial resources for this purpose;

III - reduction of regional inequalities;

IV - decentralization of science, technology and innovation activities in each sphere of government, with deconcentration in each federated entity;

V - promotion of cooperation and interaction between public entities, between the public and private sectors and between companies;

VI - stimulating innovation activity in *Instituições Científica, Tecnológica e de Inovação*, ICTs (scientific, technological and innovation institutions) and in companies, including for the attraction, constitution and installation of research, development and innovation centers and technology parks and hubs in the country;

VII - promotion of business competitiveness in national and international markets;

VIII - encouraging the creation of environments favorable to innovation and technology transfer activities;

IX - promotion and continuity of scientific and technological training and qualification processes;

X - strengthening the operational, scientific, technological and administrative capabilities of ICTs;

⁶ Id. art. 218, § 5.

⁷ Id. art. 218, § 6.

⁸ Id. art. 218, § 7.

⁹ Lei No. 10.973, de 2 de Dezembro de 2004, as amended by Lei No. 13.243, de 11 de Janeiro de 2016, art. 1, <https://perma.cc/3JRM-669G>.

XI - attractiveness of development and credit resources, as well as their permanent updating and improvement;

XII - simplification of procedures for managing science, technology and innovation projects and adoption of results-based control in their evaluation;

XIII - use of the state's purchasing power to promote innovation;

XIV - support, encouragement, and integration of independent inventors into ICT activities and the productive system.¹⁰

II. Regulation of Stem Cell Research

A. Law No. 11,105 of March 24, 2005

Article 5 of Law No. 11,105 of March 24, 2005 (Biosafety Law), states that the use of embryonic stem cells obtained from human embryos produced by in vitro fertilization and not used in that procedure is permitted for research and therapy purposes, provided the following conditions are met:

I – embryos are nonviable; or

II – embryos have been frozen for three years or more, on the date of publication of Law No. 11,105, or that have been already frozen on the date of publication of that Law, after completing three years, counted from the date of freezing.¹¹

In any case, the consent of the parents is necessary.¹² Research institutions and health services that carry out research or therapy with human embryonic stem cells must submit their projects for consideration and approval by the respective research ethics committees.¹³ The commercialization of the biological material referred to in article 5 of Law No. 11,105 is prohibited, and its practice implies the crime typified in article 15 of Law No. 9,434 of February 4, 1997.¹⁴ For the purposes of Law No. 11,105, embryonic stem cells are defined as embryonic cells that can transform into cells of any tissue in an organism.¹⁵

¹⁰ Id. art. 1, sole para.

¹¹ Lei No. 11.105 de 24 de Março de 2005, art. 5, <https://perma.cc/3C32-R4YL>.

¹² Id. art. 5, § 1.

¹³ Id. art. 5, § 2.

¹⁴ Id. art. 5, § 3.

¹⁵ Id. art. 3, XI.

B. Law No. 9,434 of February 4, 1997

According to article 15 of Law No. 9,434 of February 4, 1997, buying or selling tissues, organs or parts of the human body is punished with imprisonment, from three to eight years, and a fine.¹⁶ Anyone who promotes, mediates, facilitates, or obtains any advantage from the transaction incurs the same punishment.¹⁷

C. Decree No. 5,591 of November 22, 2005

Decree No. 5,591 of November 22, 2005, regulates provisions of Law No. 11,105, of March 24, 2005, which establishes, among other things, standards for the use, upon authorization, of embryonic stem cells obtained from human embryos produced by in vitro fertilization and not used in that procedure, for purposes of research and therapy.¹⁸

The decree further states that research institutions and health services that carry out research or therapy with human embryonic stem cells must submit their projects for consideration and approval by the respective research ethics committees, according to a resolution issued by the *Conselho Nacional de Saúde* (National Health Council).¹⁹

Article 64 of Decree No. 5591 states that the Ministry of Health is responsible for surveying and maintaining an updated registry of human embryos obtained through in vitro fertilization and not used in that procedure.²⁰ Institutions that carry out activities involving freezing and storage of human embryos must provide, in accordance with a specific standard that will establish deadlines, the data necessary to identify nonviable embryos produced in their establishments and available frozen embryos.²¹

The *Agência Nacional de Vigilância Sanitária* (ANVISA, Brazilian Health Regulatory Agency) establishes standards for procedures for the collection, processing, testing, storage, transportation, quality control and use of human embryonic stem cells for the purposes of chapter VII of Decree No. 5,591, which addresses research and therapy with stem cells obtained by in vitro fertilization.²²

According to article 66 of Decree No. 5,591, parents who donate, for research or therapy purposes, human embryonic stem cells obtained in accordance with the provisions of chapter VII of Decree No. 5,591 must sign a free and informed consent form, in accordance with specific rules of the Ministry of Health.²³

¹⁶ Lei No. 9,434, de 4 de Fevereiro de 1997, art. 15, <https://perma.cc/WP43-QU96>.

¹⁷ Id. art. 15, sole para.

¹⁸ Decreto No. 5.591, de 22 de Novembro de 2005, art. 1, <https://perma.cc/66LM-PWJQ>.

¹⁹ Id. art. 63, § 2.

²⁰ Id. art. 64.

²¹ Id. art. 64, § 1.

²² Id. art. 65.

²³ Id. art. 66.

The use, in therapy, of human embryonic stem cells, in compliance with article 63 of Decree No. 5,591 (which allows the use of embryonic stem cells for research and therapy purposes obtained from human nonviable embryos), will be carried out in accordance with the Ministry of Health guidelines for the evaluation of new technologies.²⁴

1. National Health Council - Resolution No. 466 of December 12, 2012

On December 12, 2012, the National Health Council passed Resolution No. 466, which approved guidelines and regulatory standards for research involving human beings.²⁵ The resolution incorporates, from the perspective of the individual and communities, bioethics references, such as “autonomy, non-maleficence, beneficence, justice, and equity, among others, and aims to ensure the rights and duties that concern research participants, the scientific community and the State.”²⁶ Research projects involving human beings must comply with Resolution No. 466.²⁷

The resolution states that research involving human beings must meet the relevant ethical and scientific foundations.²⁸ Respect for human dignity requires that all research be carried out with the free and informed consent of participants, individuals, or groups who, by themselves or their legal representatives, express their consent to participate in the research.²⁹ The Free and Informed Consent Process is understood as all the steps that must necessarily be observed so that persons invited to participate in research can express themselves, autonomously, consciously, freely and in an informed manner.³⁰

2. ANVISA

a. RDC Resolution No. 29 of May 12, 2008

RDC Resolution No. 29 of May 12, 2008, creates procedures relating to the Cadastramento Nacional dos Bancos de Células e Tecidos Germinativos, BCTG e Informação da Produção dos Embriões Humanos (BCTG, National Registration of Germinal Cell and Tissue Banks and Information on the Production of Human Embryos) produced by in vitro fertilization techniques and not used in that procedure.³¹

²⁴ Id. art. 67.

²⁵ Conselho Nacional de Saúde, Resolução No. 466, de 12 de Dezembro de 2012, <https://perma.cc/4C33-55VS>.

²⁶ Id. Item I.

²⁷ Id.

²⁸ Id. Item III.

²⁹ Id. Item IV.

³⁰ Id.

³¹ Agência Nacional de Vigilância Sanitária, Resolução RDC No. 29, de 12 de Maio de 2008, art. 1, <https://perma.cc/37WS-LXVX>.

The RDC Resolution states that sending information on the production of human embryos produced by in vitro fertilization techniques must be carried out by a BCTG.³²

The BCTG, a service legally constituted and linked to a health care establishment specializing in assisted human reproduction, must follow the regulation of ANVISA RDC Resolution No. 29 and that described in ANVISA RDC Resolution No. 33 of February 17, 2006, or one that replaces it.³³

Furthermore, article 3 of RDC Resolution No. 29 created the Sistema Nacional de Produção de Embriões, SisEmbryo (National Embryo Production System).³⁴

b. Resolution RDC No. 33 of February 17, 2006

Resolution RDC No. 33 of February 17, 2006, approved the technical regulations for the operation of BCTGs.³⁵ It created procedures relating a BCTG,³⁶ which is the service intended to select donors, collect, transport, register, process, store, discard and release cells and germinal tissues for therapeutic use by third parties or the donor herself.³⁷ The resolution required a service to be formally linked to a legally constituted health care establishment specializing in assisted human reproduction.³⁸

Resolution RDC No. 33 was revoked by RDC Resolution No. 23 of May 27, 2011, which provided the technical regulations for the operation of BCTGs.³⁹

On December 26, 2022, RDC Resolution No. 771 revoked RDC Resolution No. 23.⁴⁰ The new RDC Resolution provides for good practices in handling germ cells, germ tissues, and human embryos for therapeutic use in assisted human reproduction techniques, through the establishment of minimum technical-sanitary requirements related to the production cycle of these products, with a view to its safety and quality.⁴¹ The resolution applies to all establishments, whether public or private, that carry out activities with germ cells, germ tissues or human embryos for therapeutic use in assisted human reproduction techniques, whether intended for their own use or for donation.⁴²

³² Id. art. 2.

³³ Id. art. 2, sole para. ANVISA RDC Resolution No. 33 of February 17, 2006, was revoked by Resolução RDC No. 23, de 27 de Maio de 2011, <https://perma.cc/7P43-QKED>, which in turn was revoked by Resolução RDC No. 771, de 26 de Dezembro de 2022, <https://perma.cc/56WM-CJQY>.

³⁴ Agência Nacional de Vigilância Sanitária, Resolução RDC No. 29, de 12 de Maio de 2008, art. 3.

³⁵ ANVISA, Resolução RDC No. 33, de 17 de Fevereiro de 2006, <https://perma.cc/4CTR-EKJ9>.

³⁶ Id. art. 1.

³⁷ Id. art. 2.

³⁸ Id. art. 2, sole para.

³⁹ ANVISA, Resolução RDC No. 23, de 27 de Maio de 2011, <https://perma.cc/879S-96VG>.

⁴⁰ ANVISA, Resolução RDC No. 771, de 26 de Dezembro de 2022, art. 132, <https://perma.cc/7VKE-WPBW>.

⁴¹ Id. art. 1.

⁴² Id. art. 2.

c. Resolution No. 9 of March 14, 2011

On March 14, 2011, ANVISA enacted Resolution No. 9 to establish the minimum technical-sanitary requirements for the collection, processing, packaging, storage, quality control tests, disposal, release for use and transport of human cells and their derivatives aiming at safety and the quality of cells and their derivatives made available for clinical research and therapy.⁴³

Resolution No. 9 was revoked by RDC Resolution No. 214 of February 7, 2018, which provided for good practices in human cells for therapeutic use and clinical research.⁴⁴

On May 27, 2021, RDC Resolution No. 508 revoked RDC Resolution No. 214.⁴⁵ The new resolution establishes good practices in human cells for therapeutic use and clinical research.⁴⁶ According to article 2, good practices in human cells for therapeutic use and in clinical research comprise the minimum technical-sanitary requirements related to the production cycle of cells and advanced therapeutic products, with a view to the safety and quality of these products.⁴⁷ Cells or advanced therapy products that do not meet the provisions of RDC Resolution No. 508 are disqualified for therapeutic use and in clinical research.⁴⁸

3. *Ministry of Health - Administrative Act No. 2,526 of December 21, 2005*

Portaria (Ordinance) No. 2,526, enacted by the Ministry of Health on December 21, 2005, provides for the information on data necessary for the identification of human embryos produced by in vitro fertilization.⁴⁹

According to article 1, institutions that carry out activities involving the freezing and storage of human embryos produced by in vitro fertilization must inform, within 30 days counting from the publication of the rule referred to in section 2 of article 1, the data necessary to identify nonviable embryos produced in its establishments and available frozen embryos, as provided in section 1 of article 64 of Decree No. 5,591.⁵⁰ This information must be sent to the Brazilian Health Regulatory Agency for the creation of a database on human embryos.⁵¹

⁴³ ANVISA, Resolução No. 9, de 14 de Março de 2011, art. 2, <https://perma.cc/3D4Z-QA4W>.

⁴⁴ ANVISA, Resolução RDC No. 214, de 7 de Fevereiro de 2018, art. 193, <https://perma.cc/J5T2-KEBX>.

⁴⁵ ANVISA, Resolução RDC No. 508, de 27 de Maio de 2021, art. 192, <https://perma.cc/3WM2-4ZM2>.

⁴⁶ Id. art. 1.

⁴⁷ Id. art. 2.

⁴⁸ Id. art. 2, sole para. Resolução RDC No. 508, revoked.

⁴⁹ Ministério da Saúde, Portaria No. 2.526, de 21 de Dezembro de 2005, <https://perma.cc/2BU7-6EKF>.

⁵⁰ Id. art. 1.

⁵¹ Id. art. 1, § 1.

Section 2 of article 1 states that ANVISA must publish, within 120 days of the publication of Administrative Act No. 2,526, a specific rule on the data that must be reported and the forms or means for sending this data.⁵²

The free and informed consent form referred to in article 66 of Decree No. 5,591 must provide for the allocation of embryonic stem cells for research or therapy purposes and guarantee the confidentiality of the genetic data of embryo and gamete donors.⁵³

III. Developments

A. National Health Council

On April 17, 2008, the plenary of the National Health Council passed Resolution No. 400 favoring continued research with embryonic stem cells.⁵⁴ The resolution also supports the provisions of article 5 of Law No. 11,105.⁵⁵

B. Federal Supreme Court Decision

In May 2008, the Supremo Tribunal Federal (STF, Federal Supreme Court), which is the highest court in the country, authorized the scientific use of embryonic stem cells for therapeutic purposes, “thereby de-characterizing the concept of abortion and the violation of human dignity and human life itself.”⁵⁶ The high court validated article 5 of the Biosafety Law (Law 11,105/2005), which authorizes the use of embryonic stem cells obtained from human embryos produced by in vitro fertilization and not used in that procedure.⁵⁷

The decision was made in Ação Direta de Inconstitucionalidade, ADI 3510 (Direct Unconstitutionality Action No. 3510),⁵⁸ in which the Procuradoria Geral da República (Attorney General’s Office) maintained that the provision violated the constitutional protection of the right to life and the dignity of the human person, with the argument that an embryo is a human life.⁵⁹

⁵² Id. art. 1, § 2.

⁵³ Id. art. 3.

⁵⁴ Conselho Nacional de Saúde, Resolução No. 400, de 17 de Abril de 2008, art. 1, <https://perma.cc/4JVF-XFKL>.

⁵⁵ Id. art. 2.

⁵⁶ Supremo Tribunal Federal, *Mês da Mulher: STF Libera o Uso de Células-Tronco Embrionárias em Pesquisas Científicas*, <https://perma.cc/95CC-839S>.

⁵⁷ Id.

⁵⁸ Supremo Tribunal Federal, Ação Direta de Inconstitucionalidade 3.510 Distrito Federal, <https://perma.cc/JM37-6MTB>.

⁵⁹ Supremo Tribunal Federal, *Mês da Mulher: STF Libera o Uso de Células-Tronco Embrionárias em Pesquisas Científicas*, supra note 56.

The majority of the STF followed the reasoning of Minister of the Supreme Court Ayres Britto (retired), who concluded that the provisions of the Biosafety Law do not represent contempt or disregard for frozen embryos, which are unviable biologically or for the purposes for which they are intended, but rather the “valorization of fraternal constitutionalism.” The majority held that the Federal Constitution does not provide for the beginning of human life or define the precise moment at which it begins; rather, it deals with a person’s individual rights and guarantees. “Thus, a pre-implanted embryo is an asset to be protected, but not a person in the biographical sense.”⁶⁰

The decision emphasizes that the Biosafety Law does not authorize the removal of embryos from the female body. Therefore, it is not a question of termination of a pregnancy, but of an embryo resulting from an in vitro fertilization procedure, to be discarded.⁶¹

Another point highlighted in the decision is respect for the fundamental right to family planning, which involves the principles of human dignity and responsible motherhood and fatherhood. From this perspective, the option of a couple or a woman to opt for the artificial fertilization process does not entail a legal duty to use all embryos that are eventually formed and that are genetically viable. In the same sense, the process of in vitro fertilization does not imply the duty to attempt to implant all the fertilized eggs in the woman’s body (nidation). Admitting this obligation would be to treat women in an inhuman and degrading way.⁶²

Thus, for the majority, for the full right to life of the in vitro fertilized embryo to be recognized, it would be necessary to recognize the “right to a uterus, which is not authorized by the Federal Constitution.”⁶³

C. Ethics in the Use of Stem Cells

A medical article published in 2019 discusses the ethics involved in the use of stem cells and the enactment of the Biosafety Law.⁶⁴ The article highlights arguments in favor and against such use and the difficulty in determining the status of the embryo. That is, whether the embryo is a human being or not. The article affirms that the benefits derived from the use of stem cells are undeniable, but their ethical use is still an issue because the beginning of life has yet to be determined. Moreover, the indiscriminate use of such a technique could lead to a dehumanization of life, making it relatively normal to shorten life. According to the authors, these issues justified the need for the promulgation of the Biosafety Law.⁶⁵

⁶⁰ Id.

⁶¹ Id.

⁶² Id.

⁶³ Id.

⁶⁴ Danielle P. Matias et al., *A Ética no Uso de Células Tronco, Revisão de Literatura*, 25(2) UEPG Appl. Soc. Sci. 25-32 (Ponta Grossa, jul./dez. 2019), <https://perma.cc/UJ4A-XYA9>.

⁶⁵ Id.

D. Bill of Law Prohibiting the Use of Stem Cells

In 2020, Senator Eduardo Girão presented Projeto de Lei (Bill of Law), PL 5.153/2020,⁶⁶ which prohibits, for any purpose, the use of stem cells obtained from human embryos. The bill would amend the Biosafety Law, which currently allows the use of these cells for research and therapy purposes.⁶⁷

In justifying the project, the senator recognizes that adult stem cells, in medicine, represent an important therapeutic tool for serious diseases. In addition, within the scope of scientific research, they open perspectives for the development of new treatments for diseases. However, Girão expressed the need to prohibit the use of embryonic stem cells in research because, to be extracted, human embryos are destroyed and killed.⁶⁸

Girão argues further that “life begins with conception, that is, with the union of sperm and egg. Therefore, from that moment on, the embryo is a new individual with its own life and deserves to have its integrity protected. This applies equally to embryos generated through in vitro fertilization.”⁶⁹

The senator notes that the current trend in research laboratories is the use of adult stem cells, which are therapeutically effective. His bill would place no restrictions on their use, as they do not sacrifice the natural development of human embryos and can be extracted after birth until adulthood.⁷⁰

PL 5.153/2020 is currently being discussed at the *Comissão de Assuntos Sociais* (Social Affairs Committee) of the Federal Senate.⁷¹

⁶⁶ Senado Federal, Projeto de Lei No. 5.153, de 2020, <https://perma.cc/NQ7F-FBGW>.

⁶⁷ *Uso de Células-Tronco Embrionárias Pode Ser Proibido*, Agência Senado (Nov. 11, 2020, updated Dec. 07, 2020), <https://perma.cc/FH58-UC95>.

⁶⁸ Id.

⁶⁹ Id.

⁷⁰ Id.

⁷¹ Senado Federal, Projeto de Lei No. 5.153, de 2020, Tramitação.

Chile

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SUMMARY Under the Chilean Sanitary Code of 1968, a special authorization from the Public Health Institute is required for the use of pharmaceutical products and elements for scientific research on human beings. Holders of authorizations for provisional use for research purposes are liable for damages caused during the research. Law No. 20,584 of 2012, protecting the rights and duties of persons in actions related to their health care, provides that all persons have the right to grant or deny their consent to undergo any procedure or treatment related to their health care. This law strengthened the requirement of prior, informed, and written consent from patients in the case of clinical trials or any type of biomedical scientific research. The situation of impaired persons is also specially regulated so that consent may be provided by themselves or their legal representatives in the most appropriate manner. The law also allows for the establishment of Scientific Ethics Committees throughout the country to oversee and authorize scientific research on persons. In particular, Law No. 20,120 of 2006, on Scientific Research on Human Beings, Their Genomes, and Prohibiting Human Cloning contains stringent prohibitions regarding biomedical scientific research on human beings that involves some type of physical or psychological intervention, and it sets forth detailed procedures for the approval of protocols and the granting of consent by test subjects. Lastly, the Bill to Promote the Scientific Research in Cell Therapy and Stem Cells is currently pending in the Chilean Congress. The bill would amend existing legislation to establish a legal framework for the promotion of scientific research in cell therapy and stem cells.

I. Regulation of Scientific Research in General

A. The Sanitary Code of 1968

Historically, the Sanitary Code has regulated all aspects related to health in Chile, including scientific research on persons.¹

The code requires a special authorization from the Public Health Institute for the use of pharmaceutical products and elements for scientific research on human beings.² The authorization extends to any pharmaceutical product or medical device, except when its use does not entail a relevant risk for humans.³

¹ DFL No. 725 de 1968, Código Sanitario. Título V “De los Ensayos Clínicos de Productos Farmacéuticos y Elementos de Uso Médico” [Sanitary Code], <https://perma.cc/V4QB-MB27>.

² Id. art. 111A, para. 1.

³ Id. art. 111A, para. 2.

The request for a provisional authorization for research purposes must be supported by a research protocol, an informed consent form, an insurance policy, and other background information as required by the regulations.⁴

The special authorization is issued for up to one year, and it may be renewed for equal and successive periods.⁵

The Public Health Institute must keep a public record of all scientific research on human beings with pharmaceutical products or elements for medical use authorized to be carried out in the country.⁶

The authorization holder, the sponsoring entity, the principal researcher, and the respective research center must notify the Public Health Institute and the corresponding Scientific Ethics Committee of any adverse reactions and events occurred during the study.⁷ They must also show compliance with the applicable rules on pharmaco-vigilance and techno-vigilance.⁸

Patients subject to a clinical trial have a right to obtain, once the trial is completed, continuity of treatment at no cost for as long as its therapeutic usefulness persists, in accordance with the respective research protocol.⁹

Any research center where pharmaceutical products and elements for medical use in humans is carried out must be accredited by the Public Health Institute.¹⁰ The institute must also supervise compliance with research protocols, informed consents, good clinical practices, notifications of adverse reactions and adverse events and, in general, with all applicable regulations.¹¹

Holders of authorizations for provisional use for research purposes are liable for damages caused during the research, even if these arise from facts or circumstances that could not have been foreseen or avoided according to the state of scientific knowledge or technique existing at the time the damage occurred.¹² Likewise, once the damage is proven, it is presumed that it occurred during the course of the research.¹³ To that effect, holders of special authorizations for provisional

⁴ Id. art. 111A, para. 3.

⁵ Id. art. 111A, para. 4.

⁶ Id. art. 111A, para. 6.

⁷ The Comptroller General's Office has ruled that the authorization provided by a Scientific Ethics Committee for a specific scientific research project is valid even if the respective committee does not have jurisdiction over the territory where the project is to be carried out. (Dictamen No. 278 of 2019, Contraloría General de la República, <https://perma.cc/F8T9-N687>.)

⁸ Sanitary Code art. 111B.

⁹ Id. art. 111C.

¹⁰ Id. art. 111D, para. 1.

¹¹ Id. art. 111D, para. 2.

¹² Id. art. 111E, para. 1.

¹³ Id. art. 111E, para. 2.

use for research purposes of pharmaceutical products or elements for medical use must carry a civil liability insurance policy.¹⁴

B. Law No. 20,584 of 2012

The purpose of Law No. 20,584 of 2012 is to regulate the rights and duties of persons in actions related to their health care, whether health services are obtained in person, remotely, or by telemedicine.¹⁵

The basic principle is that all persons have the right to grant or deny their consent to undergo any procedure or treatment related to their health care.¹⁶ This right must be exercised freely, voluntarily, expressly, and based on adequate, sufficient and understandable information previously provided by the treating professional.¹⁷

In no case may the rejection of treatment trigger artificially accelerating death, carrying out euthanasia practices or assisting suicide.¹⁸

As a general rule, consent may be provided verbally, but a writing is required in the case of surgical interventions, invasive diagnostic and therapeutic procedures, and, in general, for the application of procedures entailing a relevant and known risk to the patient's health.¹⁹ In such cases, the acceptance or rejection must be recorded in writing in the patient's clinical record.²⁰

All children and adolescents have the right to be heard regarding treatments applied to them and to choose between alternative treatments, as the situation allows, taking into consideration their age, maturity, mental development, and emotional and psychological state.²¹ A child's or adolescent's refusal to participate or continue to participate in biomedical scientific research on humans and its clinical applications must be respected.²² If already in course, the child or adolescent must be informed of the risks of early withdrawal from treatment.²³

In the specific case of participation in any type of biomedical scientific research, all persons must be informed so they can decide on their participation in a prior, express, free, informed, personal,

¹⁴ Id. art. 111F.

¹⁵ Ley No. 20.584 de 2012, Regula los Derechos y Deberes que Tienen las Personas en Relación a su Atención en Salud [Law No. 20,584] art. 1, <https://perma.cc/WL99-RTW2>.

¹⁶ Id. art. 14, para. 1.

¹⁷ Id. art. 14, para. 2.

¹⁸ Id. art. 14, para. 3.

¹⁹ Id. art. 14, para. 4.

²⁰ Id.

²¹ Id. art. 14, para. 7.

²² Id. art. 14, para. 8.

²³ Id. art. 14, para. 9.

and written manner.²⁴ In no case may this decision mean a detriment to the person's care or any sanction.²⁵

Biomedical research may not be carried out on adults who are not physically or mentally capable of expressing their consent or whose preference cannot be known, unless the physical or mental condition that prevents them from granting informed consent or expressing their preference is a necessary characteristic of the respective human research group in which they participate.²⁶ In such cases, persons whose health condition is treatable to enable them to regain their capacity to consent may not be involved in research without consent.²⁷

In such circumstances, the research must contain specific reasons for including individuals with a disease that does not allow them to express their consent or express their preference.²⁸ Researchers must prove that the research involves a potential direct benefit for the person and implies minimal risks for them.²⁹ A previous favorable report from an accredited Scientific Ethical Committee and the respective Regional Ministerial Secretariat of Health's authorization are necessary.³⁰

Persons with neurodegenerative or psychiatric illness may give their informed consent in advance to be test subjects in future research, when it is expected that they will no longer be in a position to consent or express their preference.³¹

C. Scientific Ethics Committees

It appears that the Chilean medical scientific community views scientific research on human beings as fundamental for the development and advancement of health science and for the well-being of society.³² Likewise, a broad consensus seems to exist on the need for limits based on the essential rights and freedoms emanating from human nature, as recognized by the Chilean Constitution and international treaties ratified by Chile.³³

²⁴ Id. art. 21.

²⁵ Id.

²⁶ Id. art. 28, para. 1.

²⁷ Id.

²⁸ Id. art. 28, para. 2.

²⁹ Id.

³⁰ Id.

³¹ Id. art. 28, para. 5.

³² Natalia Herrera Medina et al., *Marco Ético y Jurídico en la Investigación Científica en Chile: Acreditación de Comités Ético Científicos* [Ethical and Legal Framework in Scientific Research from Chile: Functioning of Accredited Scientific Ethics Committees], 40(4) Int. J. Morph. 953-58 (2022), <https://perma.cc/W8CY-FYSA>.

³³ Id.

Consequently, the development of a culture of ethical conduct in medical research is desirable to establish or raise standards in scientific research ethics.³⁴ In this context, the existence of accredited Scientific Ethics Committees appears to be key.³⁵

The committees are collegiate entities constituted in public or private institutions whose essential purpose is to evaluate and report the protocols of biomedical scientific research that are submitted for their consideration.³⁶ The committees must be accredited by the Health Authority to ensure the access of their members to scientific and technical information on matters related to ethics and biomedical research and to ensure the support of resources necessary for their proper functioning.³⁷

The committees are constituted through a resolution issued by the technical director of the entity in which they are established.³⁸ A committees must issue a ruling within 45 days from the presentation of a request for medical research.³⁹ A favorable report from the committee is sufficient for the study to be carried out, provided the director of the respective establishment where the study is to be carried out also consents to it.⁴⁰

As of September 2021, there are 62 accredited committees in Chile,⁴¹ of which a majority are affiliated with a public educational or health institution.⁴²

II. Regulation of Stem Cell Research

A. Law No. 20,120 of 2006, on Scientific Research on Human Beings, Their Genomes, and Prohibiting Human Cloning

1. General Framework

Law No. 20,120 and its regulations cover all biomedical scientific research on human beings that involves some type of physical or psychological intervention.⁴³

³⁴ Id.

³⁵ Id.

³⁶ Id.

³⁷ Decreto No. 114 de 22 de noviembre de 2010, Reglamento de la Ley No. 20.120, sobre la Investigación Científica en el Ser Humano [Decree No. 114], <https://perma.cc/4FBX-7QGN>.

³⁸ Id. art. 17, para. 1.

³⁹ Id. art. 18, para. 2(2).

⁴⁰ Id. art. 18, para. 3.

⁴¹ Herrera Medina et al., *supra* note 32, at 955.

⁴² Id.

⁴³ Decree No. 114 art. 9.

The Ministry of Health and Regional Ministerial Health Secretariats are tasked with ensuring compliance with Law No. 20,120 and its regulations.⁴⁴

The Public Health Institute is in charge of authorizing the use of medicines, with or without health registration, for scientific research and clinical trials, based on a favorable report from the respective Scientific Ethics Committee.⁴⁵

A previous favorable report from a Scientific Ethics Committee accredited by the Health Authority is necessary for any scientific research on human beings to be carried out in Chile.⁴⁶ Scientific research projects to be carried out on human beings that have been authorized by a committee also need the express authorization of the director of the establishment where the projects will be carried out, and in certain cases, the Public Health Institute.⁴⁷

All serious adverse events occurring in the course of a research project must be promptly reported by the responsible researcher to the director of the establishment where the investigation is carried out and to the committee, as appropriate.⁴⁸

The genetic information of a human being must be kept confidential, without prejudice to judicial orders issued under current legislation.⁴⁹ The collection, storage, treatment, and dissemination of human genomes and all other information related to sensitive personal data emanating from a research project must comply with Law No. 19,628 on the protection of personal data and professional secrecy.⁵⁰

Human genome data that allows the identification of a person must be encrypted for storage and transmission; encryption may be temporarily omitted for reasons of public utility.⁵¹

The respective researcher is responsible for guaranteeing the safety and well-being of the participants during the course of the research as well as the scientific and ethical justification of the project and the integrity of the data collected and its respective analysis and conclusion.⁵²

⁴⁴ Id. art. 7, para. 1.

⁴⁵ Id. art. 7, para. 1(a).

⁴⁶ Id. art. 10, para. 1.

⁴⁷ Id. art. 10(bis), para. 1.

⁴⁸ Id. art. 10(bis), para. 3.

⁴⁹ Id. art. 23, para. 1.

⁵⁰ Id.

⁵¹ Id. art. 23, para. 2.

⁵² Id. art. 26, para. 1.

2. *Fundamental Principles*

Law No. 20,120 aims to protect the life of human beings from the moment of conception, their physical and mental integrity, and their diversity and genetic identity in relation to biomedical scientific research and its clinical applications.⁵³

The freedom to carry out biomedical scientific research activities on human beings is limited by respect for the essential rights and freedoms that emanate from human nature, as recognized by the Constitution and international treaties ratified by Chile.⁵⁴

Knowledge of the human genome is the common heritage of humanity. Consequently, Law No. 20,120 states that nobody may attribute or constitute ownership over it or any part of it.⁵⁵ Knowledge of the structure of a gene and the total or partial DNA sequences are not patentable.⁵⁶ However, biotechnological processes derived from knowledge of the human genome, as well as products obtained directly from them, whether diagnostic or therapeutic, are patentable according to general rules.⁵⁷

3. *Definitions*

Law No. 20,120 contains definitions of events, conduct, and stakeholders involved in medical and biological research projects involving human beings.

- Sponsor or promoter: Any individual, institution, company or organization with a domicile and legal representative in Chile, who takes the initiative to carry out and finance biomedical research—or who without directly financing the biomedical research in question has carried out specific actions to obtain financing—on human beings, and who assumes the responsibilities derived from the research established by the current legal system.⁵⁸
- Contract Research Organization: A natural or legal person, commercial, academic or otherwise, who is hired as an intermediary by the sponsor, to perform one or more of the sponsor's tasks and functions related to the clinical trial.⁵⁹
- Responsible researcher: The person who is in charge of conducting biomedical research on human beings at a site or establishment, and who assumes the responsibilities established in the law.⁶⁰

⁵³ Ley No. 20.120 de 2006. Sobre la Investigación Científica en el Ser Humano, su Genoma, y Prohíbe la Clonación Humana [Law No. 20,120] art. 1, <https://perma.cc/W76D-JVHE>.

⁵⁴ Id. art. 2 and Decree No. 114 art. 2.

⁵⁵ Law No. 20,120 art. 8, para. 1.

⁵⁶ Id.

⁵⁷ Id. art. 8, para. 2.

⁵⁸ Decree No. 114 art. 7, para. 1(b).

⁵⁹ Id. art. 7, para. 1(b.1).

⁶⁰ Id. art. 7, para. 1(c).

- Principal investigator: The researcher who is chosen from among all the researchers responsible for a multicenter study to represent them before the responsible committee.⁶¹
- Research establishment or site: The physical place or geographic space that is where the researcher plans to contact potential participants or where the intervention is carried out.⁶²
- Director of the establishment: The person who is in charge of the higher administration, direction, organization, and control of the establishment where the research is conducted.⁶³
- Multicenter research or study: The biomedical research that is conducted according to a single protocol involving two or more establishments or research sites.⁶⁴

4. *Prohibitions*

Law No. 20,120 includes the following prohibitions related to research on human beings:

- All eugenic practices (understood as the application of the biological laws of heredity to the perfection of the human species)⁶⁵ are prohibited.⁶⁶ However, the law permits genetic counseling or guidance provided by a health professional to persons with an increased risk of having offspring with a specific genetic disorder, including provision of information regarding the probability of having offspring with the disease in question.⁶⁷
- All forms of arbitrary discrimination based on a person's genetic heritage are prohibited.⁶⁸ Consequently, the results of genetic tests and predictive analyzes of the same nature cannot be used for such purposes.⁶⁹
- The cloning of human beings is prohibited, whatever the purpose pursued and the technique used.⁷⁰ Cloning of human beings is understood as the asexual generation of an organism or set of cells genetically identical to its ancestor.⁷¹
- Requesting, receiving, investigating, possessing, and using information about a genome relating to a person is prohibited, unless expressly authorized by that person.⁷²

⁶¹ Id. art. 7, para. 1(d).

⁶² Id. art. 7, para. 1(e).

⁶³ Id. art. 7, para. 1(f).

⁶⁴ Id. art. 7, para. 1(g).

⁶⁵ Id. art. 3, para. 1.

⁶⁶ Law No. 20,120 art. 3.

⁶⁷ Id. and Decree No. 114 art. 3, para. 2.

⁶⁸ Law No. 20,120 art. 4, para. 1.

⁶⁹ Id. art. 4, para. 2.

⁷⁰ Id. art. 5, para. 1 and Decree No. 114 art. 4, para. 1.

⁷¹ Decree No. 114 art. 4, para. 2.

⁷² Law No. 20,120 art. 6.

5. *Authorized Practices*

Law No. 20,120 lists authorized practices:

- The culture of tissues and organs is authorized for therapeutic diagnostic or scientific research purposes, but in no case may human embryos be destroyed to obtain the stem cells that give rise to such tissues and organs.⁷³ Those keeping samples or biological elements for research, diagnosis or treatment purposes must adopt all necessary measures for their security and confidentiality that are sufficient to protect their correct use, and must use them exclusively for the purposes for which they were obtained.⁷⁴
- Gene therapy in somatic cells is authorized only for the purpose of treating diseases or preventing their appearance.⁷⁵ Gene therapy in somatic cells is considered to be the process by which genetic material is inserted into a somatic cell so that it produces a normal protein, and its practice will be authorized only for the purposes of treatment and prevention of diseases.⁷⁶
- The genetic identity of a human being can only be investigated and determined with the person's prior and informed consent.⁷⁷
- All scientific research on human beings involving some type of physical or psychological intervention must always be carried out by qualified professionals in the field, be justified in its objective and methodology, and be in compliance in all respects with legal provisions.⁷⁸
- Scientific research cannot be carried out if there is evidence that suggests that there is a risk of destruction, death or serious and lasting bodily injury to a human being.⁷⁹
- All biomedical scientific research must have the express authorization of the director of the establishment within which it is carried out, with a prior favorable report from the corresponding Scientific Ethics Committee.⁸⁰

6. *Previous Informed Consent from Patients*

As already stated, all scientific research on a human being must have the prior, express, free, and informed consent of the patient or the patient's legal representative.⁸¹

⁷³ Id. art. 4, para. 1, and Decreto No. 114 art. 6, para. 1.

⁷⁴ Decreto No. 114 art. 6, para. 2.

⁷⁵ Law No. 20,120 art. 7.

⁷⁶ Decree No. 114 art. 5.

⁷⁷ Law No. 20,120 art. 9.

⁷⁸ Id. art. 10, para. 1.

⁷⁹ Id. art. 10, para. 2.

⁸⁰ Id. art. 10, para. 3.

⁸¹ Id. art. 11, para. 1.

Informed consent exists when the person who must provide it knows the essential aspects of the research, especially its purpose, benefits, risks, and alternative procedures or treatments.⁸²

To that effect, a patient must be provided with adequate, sufficient, and understandable information about the treatment.⁸³ In particular, patients must be duly informed of their right not to authorize the research or to revoke their consent at any time and by any means, without any responsibility, sanction or loss of benefit.⁸⁴

The consent must be recorded in a document signed by the patient or the patient's representative, the research director, and the director of the respective establishment where the treatment is to be carried out.⁸⁵

In any case, consent must be requested again each time the terms or conditions under which the research is carried out undergo modifications, unless these are considered minor by the respective Scientific Ethics Committee that has approved the research project.⁸⁶

The participation of minors in studies must not violate their essential rights or constitute a risk to their health and physical and mental integrity.⁸⁷ A minor's refusal to participate or continue in the research must be respected.⁸⁸

7. *National Bioethics Commission*

The law creates the National Bioethics Commission, composed of nine professionals and experts in bioethics appointed by the president of the republic, with the consent of the Senate.⁸⁹ The members of the commission remain in office for four years and may be re-elected.⁹⁰ The commission has an executive secretariat, which is made up of the staff assigned by the Health Ministry.⁹¹

The commission's functions include advising government agencies on ethical issues arising from scientific and technological advances in biomedicine and on matters related to biomedical scientific research on human beings. It is tasked with making recommendations on the issuance, modification, and revocation of applicable regulations.⁹²

⁸² Id. art. 11, para. 2.

⁸³ Id. art. 11, para. 3.

⁸⁴ Id.

⁸⁵ Id. art. 11, para. 4, and Decree No. 114 art. 11, para. 2.

⁸⁶ Law No. 20,120 art. 11, para. 5, and Decreto No. 114 art. 14.

⁸⁷ Decreto No. 114 art. 11, para. 1.

⁸⁸ Id.

⁸⁹ Law No. 20,120 art. 15, para. 1.

⁹⁰ Id. art. 15, para. 2, and Decree No. 114 art. 20, para. 1.

⁹¹ Law No. 20,120 art. 15, para. 3.

⁹² Id. art. 16, para. 1, and Decree No. 114 art. 19, para. 1.

Under the law, the Health Ministry may also establish a Ministerial Commission on Health Research Ethics to advise on bioethics matters, without prejudicing the functions of the Scientific Ethics Committees.⁹³

8. *Penalties*

The law punishes, among others, those who

- clone or initiate a process of cloning human beings or carry out any eugenic procedure in violation of the law,⁹⁴
- violate the confidentiality of information on human genomes, outside of the cases allowed by the law,⁹⁵
- omit the encryption of genetic information required by law,⁹⁶
- falsify a genetic record,⁹⁷ or maliciously use, for any purpose, a false record,⁹⁸
- omit the genetic record or prepare a manifestly incomplete one,⁹⁹ or
- develop a biomedical scientific research project on human beings or their genomes without having the corresponding authorizations required by law.¹⁰⁰

B. Technical Norm No. 57 of 2001

In addition, Technical Norm No. 57 of 2001, regulates the following aspects related to research involving human beings:

- informed consent of the person participating in clinical studies with pharmacological agents in human beings, setting forth the guidelines to obtain such consent,¹⁰¹

⁹³ Law No. 20,120 art. 22.

⁹⁴ Id. art. 17, para. 1.

⁹⁵ Id. art. 18, para. 1.

⁹⁶ Id. art. 18, para. 2.

⁹⁷ Id. art. 19, para. 1.

⁹⁸ Id. art. 19, para. 2.

⁹⁹ Id. art. 19, para. 3.

¹⁰⁰ Id. art. 20.

¹⁰¹ Norma Técnica No. 57, Regulación de la Ejecución de Ensayos Clínicos que Utilizan Productos Farmacéuticos en Seres Humanos (Ministerio de Salud, 2001) [Technical Norm No. 57], § III(1)(1.1.), <https://perma.cc/48QS-Z3P2>.

- commitment of the responsible researcher to the established ethical guidelines,¹⁰² which include the need for sufficient guarantees for adverse effects derived directly from the research,¹⁰³
- reports on the potential benefits to voluntary participants, which includes the development of a protocol on the basic criteria for defining a healthy individual or, when applicable, explaining a compassionate use situation,¹⁰⁴
- reports on the number of patients on whom a specific product has been previously tested and the countries in which such clinical trials have been carried out,¹⁰⁵
- existence of an establishment that meets basic infrastructure and personnel requirements adequate to carry out the studies,¹⁰⁶
- reports from the corresponding Ethical Scientific Committee that approved the study,¹⁰⁷
- requirements to be met by researchers in clinical studies with pharmacological agents in human beings,¹⁰⁸ and
- requirements to be met by sponsors of clinical studies with pharmacological agents in human beings.¹⁰⁹

III. Developments

A. General Comments

Law 20,120 is considered favorable to the development of scientific research in the country.¹¹⁰ However, the law has been the subject of numerous criticisms from the national scientific and bioethics community.

¹⁰² Id. § III(1)(1.2.).

¹⁰³ Id. § III(1)(1.3.).

¹⁰⁴ Id. § III(1)(1.5.).

¹⁰⁵ Id. § III(1)(1.9.).

¹⁰⁶ Id. § III(1)(1.10.).

¹⁰⁷ Id. § III(1)(1.11.).

¹⁰⁸ Id. § IV(1)(1.1.).

¹⁰⁹ Id. § V(1)(1.).

¹¹⁰ Bernardo Aguilera et al., *Quince Años de la Ley de Investigación con Seres Humanos: Desafíos Pendientes* [Fifteen Years of the Human Research Law: Pending Challenges], 149 Rev. Méd. Chile 1517-18 (2021), <https://perma.cc/6MZP-6TH5>.

1. *Scope of Law No. 20,120*

Critics of the biomedical approach of Law No. 20,120 argue it excludes research in the social sciences and humanities from the ethical review requirement.¹¹¹ All research involving human subjects carries risks of harm and exploitation, and the potential harm includes physical, psychological, legal, moral, and economic damage, to which the subjects participating in the research and the communities where they are located may be exposed.¹¹² For these reasons, they assert that it is necessary to expand the scope of application of Law 20,120 to cover all research with human beings.¹¹³

2. *Formal Requirements of Informed Consent*

Another criticism is that informed consent requirements should be reviewed and updated to make them compatible with diverse and emerging research methods, some of which may require obtaining consent in virtual environments, including “other ways of documenting consent, for example, through oral support, or even requesting a waiver of consent.”¹¹⁴

3. *Current Ban on Research with Human Embryos*

Additional criticism highlights that, under the current law, human embryos may not be destroyed to obtain the stem cells that give rise to human tissues and organs.¹¹⁵ However, in practice, this prohibition contrasts with the legal authorization to destroy embryos already implanted established in Law No. 21,030. Consequently, a harmonization of Law No. 20,120 and No. 21,030 is advocated.¹¹⁶

By the same token, critics assert that, along with prohibiting cloning, it would be advisable to expressly prohibit the genetic editing of germ cells, including human embryos, for reproductive purposes.¹¹⁷

4. *Unregulated Topics*

Lastly, critics argue Law No. 20,120 fails to regulate certain matters, including: the use of clinical record data for retrospective studies, the storage and use of biobank material, research with vulnerable populations or with noncompetent subjects, and conflicts of interest.¹¹⁸

¹¹¹ Id.

¹¹² Id.

¹¹³ Id.

¹¹⁴ Id.

¹¹⁵ Id.

¹¹⁶ Id.

¹¹⁷ Id.

¹¹⁸ Id.

B. Legislative Bill to Promote Scientific Research in Cell Therapy and Stem Cells

A bill pending in the Chilean Congress would amend the Sanitary Code to establish a legal framework for the promotion of scientific research in cell therapy and stem cells.¹¹⁹

1. Definitions

The bill lists several definitions relevant for the topic of stem cell research, including

- hematopoietic stem or progenitor cells,¹²⁰
- scientific research,¹²¹
- organ,¹²²
- therapeutic,¹²³
- cellular therapy,¹²⁴ and
- therapeutic products.¹²⁵

The bill also defines a clinical trial as a clinical study that meets any of the following conditions:

- a specific therapeutic strategy is assigned to the test subject or patient in advance of the trial that is not part of the usual clinical practice in Chile,
- the decision to prescribe the therapeutic products under investigation is made together with the decision to include the subject in the clinical study, or
- diagnostic and monitoring procedures or protocols are applied to test subjects or patients that go beyond usual clinical practice.¹²⁶

2. Authorized Practices

The bill would amend the Sanitary Code to

- require the authorization of the Health Services concerning processes and products that incorporate stem cells, human cells and tissues or their derivatives through allogeneic or

¹¹⁹ Boletín No. 11230-11 (Congreso de Chile, 2017), Modifica el Código Sanitario en el sentido de Establecer un Marco Legal para Fomentar la Investigación Científica en Materia de Células Madre y Terapia Celular [Stem Cell Bill], <https://perma.cc/AXC9-UNG2>.

¹²⁰ Id. art. 1(1).

¹²¹ Id. art. 1(2).

¹²² Id. art. 1(3).

¹²³ Id. art. 1(4).

¹²⁴ Id. art. 1(5).

¹²⁵ Id. art. 1(6).

¹²⁶ Id. art. 1(7).

autologous transplants in human patients, either in a clinical trial or as cell therapy that has authorizations or permits granted by the Health Services,¹²⁷

- add consideration of the special situation of vulnerable groups, such as people with legal incapacity, whether absolute or relative, and people with disabilities who may become test subjects during the process of authorizing research projects,¹²⁸
- reiterate that clinical trials and scientific research may only be authorized and carried out when they protect the rights, safety, dignity and well-being of the test subjects or patients, their interests, their health, well-being and quality of life,¹²⁹
- insist that research projects must protect the personal data of participants, their information arising from their clinical record, studies, treatments, and clinical trials,¹³⁰
- provide that the data or results obtained be supported, reliable, recorded and periodically reported to the health authority during the development of the clinical trial or scientific research in accordance with the law,¹³¹
- establish that clinical trials must be implemented free of charge for the test subject or patient, and under no undue influence or pressure on persons, with the exception of the reimbursement of expenses and productivity losses derived from the person's participation in the trial, and compensation for the inconvenience derived from their participation in it, as long as such compensation does not influence the subject's decision to participate in the study,¹³²
- indicate that, in the case of patients and possible test subjects under legal, relative or absolute incapacity, among other aspects, their withdrawal of consent must not affect the treatment already carried out based on their previously granted consent,¹³³ and
- provide that when a person with a disability has granted informed consent, the information must be provided in appropriate formats, to be accessible and understandable.¹³⁴

In addition, the bill presumes that, unless proven otherwise, the damages affecting the health of the test subject or patient during the conduct of the clinical trial or investigation and in the three years following the end of the treatment have occurred as a consequence of the clinical trial or investigation.¹³⁵ However, once that period has expired, test subjects or patients will be obliged to prove the causal link between the trial or research and the damage caused to their health.¹³⁶

¹²⁷ Id. art. 2(1), amending the Sanitary Code, art. 111.

¹²⁸ Id. art. 2(3), amending the Sanitary Code, art. 111A, para. 3.

¹²⁹ Id. art. 2(4), adding a final paragraph to the Sanitary Code, art. 111A.

¹³⁰ Id.

¹³¹ Id.

¹³² Id.

¹³³ Id.

¹³⁴ Id.

¹³⁵ Id. art. 2(6), substituting art. 111E, para. 2 of the Sanitary Code.

¹³⁶ Id.

The test subject or patient must be compensated for all possible damages and losses suffered as a consequence of the clinical trial or research and consequential damage and loss of profits derived from the impairment in the health or physical state of the person subjected to the clinical trial or scientific research, provided that this is not inherent to the pathology that is the object of the trial or investigation or to the evolution of the illness as a consequence of the ineffectiveness of the treatment.¹³⁷

In addition, placentas, organs, stem cells, and other cells, fluids, and tissues as determined by the regulations may be used for the production of therapeutic products, scientific research, clinical trials, allogeneic and autologous transplants, academic activities, import and export and other uses that the regulations indicate.¹³⁸

Finally, the activities of collection, obtaining, use or exploitation, scientific research, evaluation, manufacturing, processing, storage, distribution, administration or others related to clinical trials with placentas, organs, stem cells and other cells, fluids, and tissues from a living donor or a cadaver for scientific or therapeutic purposes will also be governed by the provisions of Law No. 19,451 of 1996, Establishing Standards on Transplantation and Donation of Organs.¹³⁹

3. Implementing Regulations

Per the bill, within two years, a regulation is to be issued by the Health Ministry to establish the program allowing distribution of homogeneous samples from the reserves of donated stem cells or blood samples for stem cells donated in the various hospitals, centers or hospital units and clinics with the capacity to generate donors, that comply with all legal requirements, among the various research or study centers on stem cells and cellular therapy in the different medical faculties of the country's universities and, in the absence of them, to the public and private research centers registered with the health authority.¹⁴⁰

4. Status of the Stem Cell Bill

The bill has been under review by the Health Commission of the Chilean Chamber of Deputies since May 16, 2017.¹⁴¹

¹³⁷ Id. art. 2(8), making an addition to art. 111F, para. 1 of the Sanitary Code.

¹³⁸ Id. art. 2(9), substituting 153 of the Sanitary Code.

¹³⁹ Id.

¹⁴⁰ Id. art. 3.

¹⁴¹ Senado, Boletín 11230-11, Modifica el Código Sanitario en el Sentido de Establecer un Marco Legal para Fomentar la Investigación Científica en Materia de Células Madre y Terapia Celular, <https://perma.cc/F238-M8TK>.

Costa Rica

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SUMMARY In Costa Rica, current scientific research legislation includes Law No. 9234 of 2014 and Law No. 40884 of 2018. These laws emphasize the importance of ethical standards and the protection of participants rights in biomedical research, including informed consent and oversight by ethical and scientific committees.

Until 2010, stem cell therapy in Costa Rica was prohibited due to its experimental nature. However, subsequent legal development introduced changes. The primary regulation on stem cell research is Law No. 39986-S of 2018, which authorized adult stem cell treatments for regenerative purposes, with specific criteria and conditions outlined. In a 2011 decision by the Supreme Court, the Court acknowledged that the petitioners had experienced no adverse effects during the treatment with stem cells, thus invalidating any legal basis for the Ministry of Health to halt ongoing treatments, allowing the plaintiffs to continue their therapy. However, this decision had limitations as it did not overturn the ban entirely. Instead, it only granted authorization to those who were currently undergoing stem cell treatment.

Recent developments in stem cell research in Costa Rica are focused on adult stem cell therapy for regenerative purposes and its potential in cancer treatment, as well as ethical concerns surrounding stem cell research and the need for bioethics considerations. There are many institutions involved in stem cell research such as the CONIS, CECs, and public and private research institutes.

Lastly, there are challenges associated with research in this area for Costa Rica that encompass ethical, legal, and logistical issues surrounding the implementation and regulation of these type of therapeutical treatments. In addition, legislation such as Law No. 9234 of 2014 focuses on biomedical studies, neglecting other forms of human research. Costa Rica lacks the infrastructure to fully comply with international ethical standards and national regulation.

I. Introduction

Costa Rica's legal system is based on the civil law tradition. The country's legal framework has evolved over the years. One of the most significant milestones in the development of the legal system was the adoption of the 1949 constitution, which established a democratic republic with a strong emphasis on protecting human rights and promoting social justice.¹

* The information in this report was provided by Stephania Alvarez, a legal research fellow currently working at the Law Library.

¹ Constitution of Costa Rica, Nov. 7, 1949, art. 1, <https://perma.cc/XVU7-XSCS> (in Spanish).

According to Costa Rica's legal tradition, a decree is a legal instrument issued by the executive branch of the government that has the force of law.² Decrees are used to regulate specific matters within the scope of the executive's authority. Unlike laws, decrees do not require the approval of the legislature and their scope is limited. They are often employed to implement laws, provide detailed regulations, or address urgent situations that require immediate action. However, they are subordinate to the constitution and laws enacted by the legislative branch.³

On the other hand, a law in Costa Rica is a legislative act passed by the National Congress, the country's legislative body. Laws are comprehensive legal norms that apply to the entire territory and are binding on all individuals and entities within Costa Rica. The legislative process for enacting laws typically involves drafting, reviewing, and debating proposed bills before they are voted on by the members of Congress.⁴

It is crucial to note the distinction between decrees and laws as part of the hierarchy of legal norms in Costa Rica.⁵ The main difference between laws and decrees lies in their origin, creation process, and scope. While both have the power to regulate specific aspects of society, laws carry more significant weight and authority, as they are the result of a democratic process involving the elected representatives of people.

II. Regulation of Scientific Research in Costa Rica

The primary legal framework governing scientific research in Costa Rica consists of Law No. 9234 of 2014, commonly referred to as the "Regulatory Law for Biomedical Research,"⁶ and the Executive Decree No. 40884 of 2018.⁷

Decree No. 40884 of 2018 establishes the foundational principles and responsibilities governing the National Health Research Council (CONIS). Pursuant to this statute, the core mission of CONIS is to ensure compliance with bioethical and scientific standards in biomedical research.⁸ This objective is achieved through the establishment of regulatory measures, diligent supervision, and continuous monitoring of biomedical research activities, all aimed at safeguarding the life, health, and dignity of individuals.⁹

Law No. 9234 of 2014 outlines the comprehensive rules and regulations governing the conduct of biomedical research in the country. Its main objective is to uphold ethical standards and to maintain the integrity of medical and scientific research, while concurrently protecting the rights and safety of research participants. Key provisions within Law No. 9234 of 2014 include the

² Constitution of Costa Rica, art. 146.

³ Ministerio de Relaciones Exteriores y Culto, Dictamen C-015 (Jan. 18, 1999), <https://perma.cc/8A8P-E2BS>.

⁴ Constitution of Costa Rica, arts. 123-129.

⁵ Civil Code, Sept. 28, 1887, art. 1, <https://perma.cc/ELT6-JAJS> (in Spanish).

⁶ Law No. 9234, Apr. 22, 2014, Diario Oficial [D.O.], <https://perma.cc/NAN4-SM52> (in Spanish).

⁷ Executive Decree No. 40884-S, Jan. 08, 2018, D.O., <https://perma.cc/H7QP-ESSP> (in Spanish).

⁸ *Misión y Visión*, Consejo Nacional de Investigación en Salud (CONIS), <https://perma.cc/6JT4-CUFV>.

⁹ Law No. 9234 arts. 35 & 43; Executive Decree No. 40884-S.

mandate for researchers to secure informed consent from study participants and the requirement that research projects strictly adhere to ethical and scientific principles.¹⁰

This law defines experimental, clinical, or interventional biomedical research as any scientific research in the field of health in which a preventive, diagnostic, or therapeutic intervention is applied to human subjects. These interventions serve the purpose of either discovering or verifying the clinical pharmacological or pharmacodynamics effects of an experimental device, a medical apparatus, or a clinical-surgical procedure, with the primary goal of evaluating the safety and efficacy or assessing the outcome of the intervention. Moreover, it stipulates that any reference to clinical research must be understood as referring to experimental clinical or interventional biomedical research on humans concerning their health.¹¹

Additionally, Law No. 9234 of 2014 introduces a framework for oversight by instituting a system that supervises and approves research projects through ethical and scientific committees, ensuring compliance with the appropriate and established ethical standards.¹² The law also incorporates provisions for the protection of the confidentiality of research participants' data. Research projects must be registered and authorized before commencement.¹³ Lastly, the law establishes penalties for non-compliance of its provisions.¹⁴

Furthermore, Decree No. 39061-S of 2015 provides a set of rules to regulate, control, and supervise the application of Law No. 9234 of 2014.¹⁵ It underscores that biomedical research involving human stem cells, the genome, or proteome must align with the fundamental objectives set forth in Law No. 9234, which emphasizes the direct or indirect improvement of an individual's health and the respect for human dignity. In particular, such research must be conducted in accordance with the principles outlined in the Universal Declaration on the Human Genome and Human Rights by UNESCO.¹⁶

III. Regulation of Stem Cell Therapy and Research

A. Ban on Stem Cell Therapy

In 2010, the Ministry of Health of Costa Rica imposed a prohibition on stem cell therapy due to its experimental nature. This prohibition left numerous patients unable to continue their stem cell therapy treatment.¹⁷ However, the Supreme Court of Costa Rica, in a subsequent ruling,

¹⁰ Id. art. 9.

¹¹ Id. art. 2.

¹² Id. art. 46 & 48.

¹³ Id. art. 13.

¹⁴ Id. arts. 70 – 84.

¹⁵ Executive Decree No. 39061-S, May 8, 2015, art. 1, <https://perma.cc/THY5-F8X2> (in Spanish).

¹⁶ Id. art. 5.

¹⁷ Rory Carroll, *Costa Rican Health Ministry Bans Experimental Stem Cell Treatment*, *The Guardian* (June 7, 2010), <https://perma.cc/BB3U-PVGH>.

emphasized that medical treatments must receive legal authorization prior to implementation.¹⁸ The Court acknowledged that the petitioners had experienced no adverse effects during the treatment with stem cells, thus invalidating any legal basis for the Ministry of Health to halt ongoing treatments.¹⁹ This decision was founded on the argument that the plaintiffs possessed an inherent right to human dignity and health, thereby allowing the them to continue with their therapy.²⁰ This decision is limited in nature because it did not reverse the ban, it only authorized the people who were receiving the stem cell treatment at the time to continue it under the supervision of the Ministry of Health.²¹

B. Authorization of Adult Stem Cell Therapy

In 2016, Costa Rica enacted Decree No. 39986-S of 2016,²² which authorized adult stem cell treatments for regenerative therapies while establishing the requisite criteria applicable to such treatments within the country. This law also delineates the conditions governing tissue procurement and cell isolation procedures.

Notably, Decree No. 39986-S of 2016 specifies that no authorization is necessary for progenitor cell transplants sourced from peripheral blood or hematopoietic cell transplants derived from umbilical cord blood when employed in treatment of specific medical conditions. These conditions include acute myeloid or lymphoid leukemia, chronic lymphocytic or myeloid leukemia, Hodgkin's lymphoma or non-Hodgkin's lymphoma, severe aplastic anemia, paroxysmal nocturnal hemoglobinuria, immune system diseases, hemoglobinopathies and hereditary metabolic diseases, multiple myeloma, and certain solid tumors.²³

Additionally, Decree No. 39986-S of 2016 stipulates that stem cell therapy subjected to more than minimal manipulation requires an authorization by the interested party that wants to provide this service. To obtain authorization, the interested party must submit a written application to the relevant government agency, accompanied by evidence of completed preclinical studies, demonstrating that it is effective and safe for use in clinical practice. Failure to meet this requirement mandates the procedure under a research protocol, in accordance with Law No. 9234 of 2014 and Executive Decree No. 39061-S of 2015.²⁴

Facilities offering adult stem cell therapy exclusively for autologous use aimed at regenerative purposes and exposed to minimal manipulation, will be authorized to operate this service. Nevertheless, these facilities must submit a written request to the governing government agency. The request must include a scientifically reasoned justification establishing the procedure's

¹⁸ Corte Suprema de Justicia, Sala Constitucional, Mar. 9, 2011, Resolución No. 2011003078, Expediente No. 10-005895-00007-CO, <https://perma.cc/FC5Y-D535>.

¹⁹ Id.

²⁰ Id.

²¹ Id.

²² Executive Decree No. 39986-S, Sept. 14, 2016, <https://perma.cc/KDX8-PTVP> (in Spanish).

²³ Id. art. 2.

²⁴ Id. art. 3.

reasonable chance of success, supported by preclinical proof of concept evidence attesting to its efficacy and safety. Failure to meet this requirement also demands the execution of the procedure under a research protocol, as stipulated by Law No. 9234 of 2014.²⁵

Moreover, for adult stem cell therapy for exclusive autologous use, comprehensive documentation is mandated. This includes a detailed characterization of the cell types to be transplanted, their processing and production, administration protocols (including adjuvant drugs, agents, and surgical procedures), a plan for clinical follow-up, and data records to ensure the therapy's effectiveness and monitor or prevent adverse effects. Additionally, the original certification of the stem cell therapy personnel's training must be provided, along with a copy of all these documentations.²⁶

The Executive Decree further specifies that health services offering adult stem cell therapies for regenerative purposes must obtain a certificate authorizing this service and must comply with the following conditions:

- Authorized stem cell therapy must be prescribed by a medical professional;
- There has to be informed consent, specifying that the patient that will receive the adult stem cell therapy acknowledges the possible benefits and risks of the procedure via his or her signature on the consent form;
- A report identifying adverse effects on the patient receiving the adult stem cell therapy must be sent to the National Pharmacovigilance Center of the Ministry of Health according to the Regulation on Good Pharmacovigilance Practices;²⁷
- An action plan for addressing adverse effects, encompassing their progression over time, provision of appropriate medical care, and if necessary, psychological support must be provided; and
- The initial collection of tissue must be done by using sterile techniques and basic precautions to minimize risks of contamination, infection, or pathogenic transmission. In cases where there is a more than minimal manipulation of adult stem cells, this has to be conducted within a clinical laboratory authorized by the Ministry of Health.²⁸

Lastly, for the registration and commercialization of cell-based products, stringent criteria regarding efficacy and safety in both clinical and non-clinical studies must be met, in accordance with the requirements specified in the Biomedical Research Regulatory Law²⁹ and its accompanying Statute.³⁰

²⁵ Id. art. 4.

²⁶ Id.

²⁷ Executive Decree No. 35244-S, Apr. 13, 2009, <https://perma.cc/3HCR-JYQR> (in Spanish); Executive Decree No. 39417-S, Oct. 5, 2015, <https://perma.cc/Q5A7-WWMP> (in Spanish).

²⁸ Executive Decree No. 39986-S art. 5.

²⁹ Law No. 9234.

³⁰ Executive Decree No. 39061-S.

IV. Developments of Stem Cell Research

In recent years, Costa Rica has experienced significant advancements and publications pertaining to health-related research.³¹ This includes stem cell research and its associated legislative framework although this domain is still limited in Latin American countries such as Costa Rica.³² These developments reflect the country's growing interest and involvement in the field of regenerative medicine.

A. Adult Stem Cell Therapy for Regeneration

A recent publication highlights the practice of adult stem cell therapy in Costa Rica, emphasizing its role in cellular regeneration and repair. The publication highlights the concerns that arise from unauthorized stem cell therapy treatments in Latin America, specifically when these treatments lack scientific evidence, failing to ensure patients' safety and the efficacy of the treatments. In addition, the publication specifies that to engage in regenerative treatments, a cell manufacturing certificate is required for adult stem cell regenerative treatments in Costa Rica. However, the manufacturing conditions remain unspecified according to national legislation, leading to uncertainty regarding the manufacturing process.³³

1. Cancer and Stem Cells

Recent research has focused on the relationship between cancer and stem cells due to the therapeutic potential for cancer patients.³⁴ In addition, in Latin America, there is an increase in access to hematopoietic cell transplantation.³⁵ However, publications on cancer and stem cells therapy in Costa Rica are limited.

2. Embryonic Stem Cells

In Costa Rica, it is only permitted to research on adult stem cells as specified by Law No. 39986-S of 2016. Therefore, the use of embryonic cells is not allowed; specifically due to the regulation that grants the human embryo personhood. Research with embryonic stem cells can violate the

³¹ Saray Córdoba et al., *La Producción científica indexada de América Central y República Dominicana* (2019), <https://perma.cc/37SJ-AUJ4>.

³² Tanya Estefanía Orozco-Solares et al., *Allogeneic Mesenchymal Stem Cell-Based Treatment Legislation in Latin America: The Need for Standardization in a Medical Tourism Context*, 31 *Stem Cells Dev.* 143-162 (2022), <https://perma.cc/2XNC-JH7E>.

³³ Id.

³⁴ Luis Arias-Espinosa et al., *Acute Leukemia Relapse after Hematopoietic Stem Cell Transplantation: The Good, the Bad, and the Ugly of Isolated Extramedullary Relapse in a Latin American Population*, 29 *Transplant Cell Ther.* 510e1-510e9 (2023), <https://perma.cc/FM5L-CMLN>; Miguel Angel Chiurillo, *Role of Polymorphisms in Gastric Cancer and its Precursor Lesions: Current Knowledge and Perspectives in Latin American Countries*, 20 *World Journal of Gastroenterology* 4503-4515 (2014), <https://perma.cc/BK98-Q3Z3>.

³⁵ Cinthya Correa et al., *Increasing Access to Hematopoietic Cell Transplantation in Latin America*, 57 *Bone Marrow Transplant* 881-888 (2022), <https://perma.cc/Y2PU-3Q2B>.

right to life of the unborn. This shows a more restraining policy framework on human embryo research in Costa Rica.³⁶

3. Bioethics and Stem Cell Therapy

Ethical concerns surrounding the sourcing and utilization of stem cells may present significant challenges. Ensuring that research aligns with ethical principles and safeguards the rights of patients is of utmost importance. Research articles on human research in Costa Rica highlight the relevance of bioethics and human rights particularly in scientific and clinical research, such as stem cell clinical research.³⁷ Furthermore, authors have said there is a need to provide training to researchers about human research ethics and regulation.³⁸

B. Relevant Institutions in Scientific Research

The most relevant institution in Costa Rica regarding Scientific Research is CONIS. This institution was created by Law No. 9234 of 2014.³⁹ This independent, multidisciplinary, ethical, technical, and scientific council, attached to the Ministry of Health, seeks to guarantee the quality of biomedical research and strict adherence to human rights guidelines.⁴⁰

In addition, Law No. 9234 of 2014 specifies that all public or private entities that conduct biomedical research may constitute a CEC (Comité Ético Científico) duly accredited by the CONIS.⁴¹ CECs have the responsibility for reviewing, supervising, and auditing research protocols.⁴² In Costa Rica, there are several research institutes, for example, the Costa Rican Institute of Clinical Research. The University of Costa Rica offers courses through its Office of Research with the purpose of training professionals in high quality research in health related topics.⁴³ In addition, this University has the Institute of Health Research which promotes human health with scientific investigation.⁴⁴

³⁶ Rosario M. Isasi et al., *Legal and Ethical Approaches to Stem Cell and Cloning Research: A Comparative Analysis of Policies in Latin America, Asia, and Africa*, 32 J. L. Med. & Ethics 626-640 (2004). <https://perma.cc/8D2S-7S8Q>.

³⁷ Francisco Quesada Rodriguez, *La bioética y los derechos humanos: una perspectiva filosófica sobre la justicia en la investigación científica y experimentación clínica con seres humanos*, 30 Med. Leg. Costa Rica 24-34 (2013), <https://perma.cc/988H-LZPD>; Michael Householder et al., *Reviving Human Research in Costa Rica* 41, J. Ethics & Hum. Rsch., (Feb. 2019), <https://perma.cc/5PY4-AJM7>.

³⁸ Michael Householder et al., *supra* note 37.

³⁹ Law No. 9234 art. 34.

⁴⁰ Id. art. 34 and 35.

⁴¹ Id. art. 46.

⁴² Id.; *Quiénes somos*, Consejo Nacional de Investigación en Salud (CONIS), <https://perma.cc/6ZH3-U45J>.

⁴³ Michael Householder et al., *supra* note 37.

⁴⁴ *Objetivos*, Universidad de Costa Rica-Instituto de Investigaciones en Salud (INISA), <https://perma.cc/39U5-BF6F>.

Another relevant institution is El Colegio de Médicos y Cirujanos. This medical association has stated that adult stem cell therapy requires the completion of research studies involving humans and should not be offered to patients as treatment but instead as free clinical research. Therefore, they agree that this area should be regulated in a way that permits clinical research to be efficient and safe for humans.⁴⁵

C. Critiques to Regulatory Law of Biomedical Research

Challenges in scientific research related to stem cells in Costa Rica encompass ethical, legal, and logistical issues surrounding the implementation and regulation of these types of therapeutical treatments. Law No. 9234 of 2014 has faced criticism for its focus on biomedical studies, neglecting other forms of human research. Moreover, some of its terminology is imprecise. Additionally, it emphasizes the need for training in responsible research conduct without specifying the contents or formats of such training. Costa Rica also lacks the infrastructure to fully comply with international ethical standards and the Regulatory Law of Biomedical Research or Law No. 9234.⁴⁶

⁴⁵ Carlos Arrea B., Oscar Porras & María Paz León-Bratti, *Las células madre y su uso en seres humanos*, 52 Acta Méd. Costarric. (2010), <https://perma.cc/4QUN-3PN9>.

⁴⁶ Michael Householder et al., *supra* note 37.

El Salvador

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SUMMARY El Salvador has regulated scientific research mainly through Decree No. 234 of 2013, which sets the guidelines for the development of science and technology. This framework includes the formulation of a national policy and plan, contributing to the country's National Research Agenda.

Additional guidance is found in the National Policy on Health Research and Decree No. 307 of 2016, which establishes rules pertaining to the administration of informed consent for medical research.

The country's regulation of stem cell research is governed by the Law on the Transplant of Human Cells, Tissues, and Organs. This comprehensive legal framework provides relevant principles and protocols for the collection and transplantation of human cells, tissues, and organs. Additionally, it strictly prohibits the use of human blood, cells, and tissues for reproductive purposes. Furthermore, the Ministry of Health and the Superior Council of Transplants (CST) are relevant institutions that oversee healthcare institutions and ensure compliance with this regulation. Non-compliance with the regulation can result in strict sanctions.

Furthermore, non-compliance with the established regulations may result in penalties, highlighting the importance of adhering to ethical and legal standards in scientific research. However, the sector still faces challenges, such as the absence of specific laws governing biomedicine research, inadequate institutional budgets, and a scarcity of skilled human resources, underscoring the need for comprehensive reforms and strategic investments in the field.

El Salvador has made significant progress in scientific research, exemplified by the notable increase in publications between 2001 and 2010. Institutions such as the University of El Salvador, the Benjamin Bloom National Children's Hospital, and the National Institute of Health (INS) have played a crucial role in advancing the country's scientific research landscape. Nonetheless, further research and resources are required to strengthen the scientific research sector in the country, particularly in the domain of stem cell research.

I. Regulation of Scientific Research in General

Throughout the last decade, El Salvador has developed a legal framework to regulate scientific research. Decree No. 234 of 2013 sets forth the parameters for the development of science and technology, by defining institutional and operational instruments and mechanisms for the

implementation of a National Policy and Plan, which jointly constitutes the frame of reference of the National Research Agenda.¹

The National Policy delineates the principles and strategies guiding scientific and technological activity to bolster the generation, utilization, dissemination, and transfer of knowledge.²

The National Plan serves as the planning tool directing government agencies to advance in science and technology, including research in these domains.³ Moreover, the Decree encompasses the establishment of a National Science, Technology, and Innovation System, comprising public and private institutions, businesses, universities, and organizations tasked with promoting, cultivating, and supporting research, as well as fostering development and innovation across diverse fields.⁴

Under Accord No. 1704 of 2016, a National Policy on Health Research was instituted. This policy outlines the government's commitment to allocating efforts and resources toward health research through the construction of a legal framework to facilitate its implementation, operation, and objectives. Additionally, it expounds upon the definition of health research and the ethical standards of research in this domain.⁵

Furthermore, Decree No. 307 of 2016 established rules pertaining to the administration of informed consent for medical research. It stipulated that any patient participating in medical research must provide written consent and confirm that they received adequate and sufficient information.⁶

Decree No. 302 of 2019 designated the Ministry of Health, operating through the National Institute of Health (INS), as responsible for the oversight of health-related research. Their responsibility entails the application of ethical principles and the scientific method in accordance with established national and international standards, with the goal of producing and integrating knowledge that contributes to the formulation of public health policies.⁷

II. Regulation of Stem Cell Research

Stem cell research is a complex and evolving field, offering potential medical breakthroughs while also raising ethical and legal considerations.

* The information in this report was provided by Stephania Alvarez, a legal research fellow currently working at the Law Library.

¹ Decree No. 234, Feb. 15, 2013, art. 1, <https://perma.cc/Q3Q8-XK6P>.

² Id. art. 4.

³ Id. art. 6.

⁴ Id. art. 11.

⁵ Ministry of Health, *Política Nacional de Investigación para la Salud* (2017), <https://perma.cc/VAH4-4KGZ>.

⁶ Ley de Derechos y Deberes de los Pacientes, art.16, <https://perma.cc/XN3X-SUBH>.

⁷ Decree 302, art. 28, <https://perma.cc/HQU8-E75Q>.

In El Salvador, the Law on the Transplant of Human Cells, Tissues, and Organs is a comprehensive legal framework governing various facets of stem cell research. From outlining the protocols for collection and their therapeutic applications to emphasizing the importance of ethical conduct, this legislation underscores the country's commitment to responsible scientific advancement.⁸

Central to this law is the regulation on the research, donation, preservation, storage, import, transport, destination, and final disposal of anatomical components and their derivatives, inclusive of hematopoietic progenitor cells. This stringent oversight is aimed at fostering innovative research and educational initiatives while upholding the highest ethical standards within the scientific community. However, this law strictly prohibits the use of human blood and blood components, cells, and tissues for any purposes related to human reproduction, emphasizing the government's ethical stance on this issue.⁹

The Ministry of Health assumes responsibility for the national policy on donation and transplant of anatomical components and their derivatives, including hematopoietic progenitor cells, for therapeutic purposes and research.

A critical component of this regulatory landscape involves securing informed consent from donors, ensuring that individuals are fully apprised of the implications of their participation in stem cell research.¹⁰ Moreover, the law distinguishes between different types of stem cells, allowing the use of adult stem cells for therapeutic procedures while strictly prohibiting the utilization of embryonic stem cells, thus reflecting the ethical considerations central to the legal framework.¹¹

This legislation sets forth important principles such as confidentiality and altruism. Emphasizing the significance of donor information confidentiality, it safeguards the privacy of individuals who opt to donate adult stem cells or organs.¹² In addition, the regulation promotes altruism and gratuity of donation as fundamental principles of organ and tissue donation, encouraging donors to contribute to research and therapeutic efforts without expecting personal gain to help others.¹³

The decree allows the use of stem cells and organs from deceased individuals for research and therapeutic purposes, acknowledging the potential benefits of post-mortem donations.¹⁴ The decree establishes clear administrative and technical requirements for institutions involved in acquiring and implanting human tissues and operating tissue banks.¹⁵ These requirements are crucial to ensure the responsible and ethical conduct of research activities.

⁸ Decree 348, Feb. 15, 2022, <https://perma.cc/2H6Q-XJ57>.

⁹ Id. art. 1.

¹⁰ Id. art. 19.

¹¹ Id. art. 42.

¹² Id. art. 3(b).

¹³ Id. art. 3(c) & (d).

¹⁴ Id. arts. 21- 24.

¹⁵ Id. art. 35.

Researchers and medical institutions engaged in stem cell research must obtain proper authorizations and licenses from the competent authority, which ensures that research proposals are scientifically valid, ethically sound, and legally compliant.¹⁶ Therefore, this legislation has created the Superior Council of Transplants of El Salvador (CST), which serves as the competent authority responsible for granting authorizations, monitoring research activities, and ensuring compliance with ethical and legal standards.¹⁷

Furthermore, this legislation mandates healthcare establishments carrying out donation and transplantation processes **to establish a committee on donation, transplantation, and bioethics**. These committees must be authorized by the CST.¹⁸ Additionally, the CST will determine the composition and functions of these committees.¹⁹ Healthcare establishments performing hematopoietic cell procedures must have an authorization from the CST.²⁰ Failure to comply with these requirements or conducting treatments or procedures with hematopoietic progenitor cells or other human cells in treatments not scientifically approved, will result in sanctions.²¹

III. Developments

Although historically El Salvador has fallen behind in scientific research compared to other countries in the region, the nation has recently made significant strides bolstering its scientific capabilities. Recognizing the potential for scientific knowledge to drive sustainable economic and social development, the country has doubled its capacity for generating scientific knowledge, leading to a notable increase in publications.²²

According to the country's National Health Policy, the period from 2001 to 2010 witnessed a remarkable upsurge in research publications. El Salvador's average publication count surged from 48 in 2001 to 152 in 2010.²³ This increase in scientific production reflects the country's dedication to enhancing its research landscape. It is important to note that compared to its regional counterparts, El Salvador, along with Costa Rica, has displayed comparatively lower levels of research with international collaboration.²⁴ Analyzing the leading institutions in terms of scientific production, the University of El Salvador emerges as a frontrunner, contributing to 60 publications (10.3%) during this lapse of time, followed by the Benjamin Bloom National Children's Hospital with 35 (6%).²⁵ In addition, the National Institute of Health (INS) has

¹⁶ Id. arts. 12(e), 26, 31 & 34-36.

¹⁷ Id. art. 7.

¹⁸ Id. art. 33.

¹⁹ Id. arts. 48 & 50.

²⁰ Id. art. 26.

²¹ Id. arts. 49 & 50.

²² Ministry of Health, *supra* note 5, at 16.

²³ Id.

²⁴ Id.

²⁵ Id.

increased its publication output, playing a pivotal role in advancing the country's scientific research landscape.²⁶

Despite these advancements, the domain of scientific publications concerning stem cells research in El Salvador remains comparatively limited. Critics of scientific health research in the country have pinpointed the absence of a specific law governing biomedicine research, the inadequate allocation of institutional budgets for research development, and a shortage of human resources possession the requisite technical expertise essential for robust research advancement.²⁷ They argue these limitations underscore the pressing need for comprehensive reforms and strategic investment in the field of scientific research concerning stem cell research within El Salvador.²⁸

²⁶ Id. at 17.

²⁷ Dalia Xochitl Sandoval, *Estado de la Investigación en Salud en El Salvador* (2008), <https://perma.cc/EF42-CRWP>.

²⁸ Id.

Mexico

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SUMMARY Health research in Mexico involving human beings is subject to several requirements, including informed consent from participants in such projects, which must be conducted by health professionals in medical institutions. Mexico's Department of Health has published a proposal in order to receive public comments on guidelines addressing a number of therapeutic matters pertaining to stem cells, which include a section on stem cell research. The Mexican government has indicated that one goal of the proposed regulation is to prohibit the false advertising of cures for diseases based on unproven cell-based treatments that have been publicized and administered in Mexico. As of October 2023, this proposed regulation has not been approved.

I. Regulation of Scientific Research in General

Mexico's Constitution provides that the government must support scientific, humanistic, and technological research and innovation and guarantee open access to the information derived from it, for which it must provide sufficient resources and incentives, in accordance with pertinent guidelines established by relevant statutes.¹

Accordingly, Mexico's Health Act provides that the Department of Health (DOH) must support the operation of institutions dedicated to health research, particularly on the following topics:

- biological and psychological processes in human beings,
- prevention and control of health problems that are considered a priority for the population,
- control of the harmful effects of the environment on health,
- study of techniques and methods recommended or used for the provision of health services, and
- domestic production of health products and supplies.²

In general, health research involving human beings is subject to several requirements, including the following:

¹ Constitución Política de los Estados Unidos Mexicanos, as amended, art. 3-V, Diario Oficial de la Federación [D.O.F.], Feb. 5, 1917, <https://perma.cc/D7FJ-37WV>.

² Ley General de Salud, as amended through 2023, arts. 96, 97, D.O.F., Feb. 7, 1984, <https://perma.cc/SSN7-2MCY>.

- It may be carried out only when the knowledge to be produced cannot be obtained by any other suitable method, and when there is reasonable assurance that it does not expose the participants to unnecessary risks or harm.
- It may only be performed by health professionals in medical institutions.
- Written consent must be obtained from the participants or their legal representatives if participants lack legal capacity, after they are made aware of the objectives of the experiment and of the possible positive or negative consequences for the participants' health.³

In order for informed consent to be deemed valid, the participants or their legal representatives must receive a clear and complete explanation of a number of aspects of the research project in which the participants are involved, including

- the justification and objectives of the research,
- the expected risks,
- the benefits that may be obtained,
- alternative procedures that may be advantageous for the participants,
- the guarantee of receiving an answer to any questions and clarification of any doubts about the procedures, risks, benefits, and other matters related to the research project,
- the freedom to withdraw consent at any time and stop participating in the study,
- the assurance that participants will not be identified and that confidentiality will be preserved, unless they expressly agree to be identified,
- a commitment to provide the participants with up-to-date information obtained during the study, even if it may affect their willingness to continue participating,
- the availability of medical treatment and the compensation to which the participants would be legally entitled from the entity conducting the research project in the event that damages occur that are directly caused by the experiment, and
- a commitment that, if there are additional expenses, they will be covered by the research budget.⁴

II. Regulation of Stem Cell Research

The DOH has the authority to issue technical regulations governing research procedures on health matters.⁵ Accordingly, the DOH has published a proposal in order to receive public comments on guidelines addressing a number of therapeutic matters pertaining to stem cells,

³ Id. art. 100.

⁴ Reglamento de la Ley General de Salud en Materia de Investigación para la Salud, as amended, arts. 16, 20, & 21, D.O.F., Jan. 6, 1987, <https://perma.cc/62SS-9JZ7>.

⁵ Id. arts. 2, 5-A(I). See also Ley General de Salud art. 17-bis.

which include a section on stem cell research.⁶ Per the proposal, such research would be subject to a number of requirements, including the following:

- The research must have authorization from DOH, which is contingent on compliance with applicable criteria for health research projects.
- The establishments that carry out the research must be accredited by health authorities.
- The lead researcher must have relevant medical expertise and professional licenses approved by Mexican authorities.
- The dissemination of information on, or advertisement of, the therapeutic use of regenerative medicine or cell therapies that have not been scientifically proven is prohibited.⁷

III. Developments

The Mexican government has indicated that one of the goals of the proposed regulation on the therapeutic uses of stem cells is to prohibit the false advertising of cures for many types of diseases based on unproven cell-based treatments that have been publicized and administered in Mexico.⁸ As of October 2023, this proposed regulation has not been approved.

⁶ Proyecto de Norma Oficial Mexicana PROY-NOM-260-SSA1-2017 para la Disposición de Células Troncales y Progenitoras con Fines Terapéuticos y de Investigación, § 20, Investigación con Células Troncales y Progenitoras, Mar. 2, 2017, <https://perma.cc/ZH6Z-MWZD>.

⁷ Id.

⁸ Presione Soltar, Secretaría de Salud, Uso de Células Madre Contará con Marco Regulatorio: Esta Herramienta Busca Terminar con Publicidad Engañosa y Mal Uso de Células Madre (Aug. 19, 2018), <https://perma.cc/XS9E-AWUP>.

Panama

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SUMMARY Panamanian law provides that the Ministry of Health (MOH) has the authority to establish policies aimed at guiding and promoting research on health matters and to ensure that such guidelines are adhered to by relevant entities, particularly with respect to human clinical trials, which have to be authorized by health authorities before they begin. Research on tissues and cells of human origin must take place in health establishments duly licensed to conduct transplants of hematopoietic tissue. The use of cells and tissue of human origin for clinical research in humans is to be evaluated by MOH in accordance with applicable regulations on clinical research and relevant international guidelines. A bill that would establish guidelines for the cultivation, development, reproduction, and administration of mesenchymal stem cells has been introduced in Panama's legislature.

I. Regulation of Scientific Research in General

Panamanian law provides that the Ministry of Health (MOH) has the authority to establish policies aimed at guiding and promoting research on health matters and to ensure that such guidelines are adhered to by relevant entities, particularly with respect to human clinical trials, which have to be authorized by health authorities before they begin.¹

Authorization for these projects may be granted provided that the requests comply with applicable requirements, which include a proposal indicating the protocols to be followed, the participating institutions and researchers, and the funding sources.²

Health authorities are to implement mechanisms to monitor and evaluate research projects on health matters, and the directors of the health institutions and establishments where these projects take place must ensure compliance with applicable guidelines.³ Researchers in charge of the projects must report to health authorities the findings of trials, particularly if they may have a significant effect on public health.⁴

¹ Ley 84 de 2019 que Regula y Promueve la Investigación para la Salud y Establece Su Rectoría y Gobernanza, y Dicta Otras Disposiciones arts. 1, 10, 11, 35, & 37, Gaceta Oficial [G.O.], May 16, 2019, <https://perma.cc/L76V-AXEG>.

² Id. arts. 45, 46, 49, 50, 54, & 58.

³ Id. art. 58.

⁴ Id. art. 61.

II. Regulation of Stem Cell Research

Panamanian law provides that research on tissues and cells of human origin must take place in health establishments duly licensed to conduct transplants of hematopoietic tissue.⁵ The use of cells and tissue of human origin for clinical research in humans is to be evaluated by MOH in accordance with applicable regulations on clinical research and relevant international guidelines, including

- the International Ethical Guidelines for Health-Related Research Involving Humans prepared by the Council for International Organizations of Medical Sciences in collaboration with the World Health Organization,
- the principles established by the Declaration of Helsinki of the World Medical Association, and
- The Standards of Good Clinical Practice derived from the International Conference on Harmonization.⁶

Informed consent must be obtained from donors of allogeneic tissue or cells (including from mothers who donate their umbilical cords).⁷ Such consent must indicate the specific purpose planned for the tissue or cells and that no compensation is involved.⁸ Human embryos may not be created for the sole purpose of obtaining stem cells from them.⁹

III. Developments

In 2020, a bill was introduced in the Panamanian legislative assembly that would establish guidelines for the cultivation, development, reproduction, and administration of mesenchymal stem cells (MSCs) in Panama.¹⁰ The preamble of the bill states that the current legal framework on stem-cells treatments is not entirely appropriate to govern MSCs-based novel techniques to treat illnesses.¹¹ More specifically, the bill states that the current legal framework considers stem cell-based treatments as “transplants,” and thus imposes strict transplant regulatory requirements on them.¹²

⁵ Decreto Ejecutivo N° 179 que Reglamenta la Investigación con Tejidos y Células de Origen Humano en la República de Panamá art. 2, G.O., June 13, 2018, <https://perma.cc/YC5P-JHU5>.

⁶ Id. art. 3.

⁷ Id. art. 6.

⁸ Id.

⁹ Id. art. 8.

¹⁰ Proyecto de Ley 179 que Establece las Guías para el Cultivo, Desarrollo, Reproducción y Aplicación de Células Mesenquimales y Sus Derivados en la República de Panamá, Jan. 7, 2020, <https://perma.cc/A9GL-RTSB>.

¹¹ Id. Exposición de Motivos.

¹² Id.

The bill proposes to exclude MSCs-based treatments from being considered as transplants, because the former may be administered through procedures that are less invasive than the latter.¹³ Furthermore, MSCs, which may help the receiving patient's body to regenerate human tissues or organs, become "inactive" in the receiving patients' bodies in about four to six months.¹⁴ This does not happen with transplants, which are designed to stay in the patients' bodies.¹⁵ Thus, the bill proposes a new regulatory regime for this type of treatment, which can be administered with less stringent requirements than those applicable to transplants.¹⁶ As of October 2023, this bill has not been approved by the Panamanian legislative assembly.

¹³ Id.

¹⁴ Id.

¹⁵ Id.

¹⁶ Id.

Regulation of Stem Cell Research: Table of Primary Sources

Jurisdiction	Type of Law	Citation	URL
Argentina	Statute	Ley No. 3.852 de 29 de julio de 2004 de la Provincia de Rio Negro	https://perma.cc/HR32-EEEP
		Ley No. 24.742 del 27 de noviembre de 1996, Sistema Público de Salud	https://perma.cc/9QKR-WMKN
		Ley No. 25.392 de 30 Noviembre de 2000, Registro Nacional de Donantes de Células Progenitoras Hematopoyéticas	https://perma.cc/U9BQ-3522
		Ley No. 25.467 de 29 de agosto de 2001, Sistema Nacional de Ciencia, Tecnología e Innovación	https://perma.cc/XC52-CH2Y
		Ley No. 26.879 de 3 de julio de 2013, Créase el Registro Nacional de Datos Genéticos	https://perma.cc/VK95-5PEL
		Ley No. 5.165 de 1 de septiembre de 2005 de la Provincia de Catamarca, Garantía y Resguardo del Derecho a la Dignidad, Identidad e Integración de Todas las Personas en Relación a su Patrimonio Genético	https://perma.cc/AE53-PTMM
		Ley No. 5.228 de 10 de octubre de 2004 de la Provincia de Chubut	https://perma.cc/9H24-62EA
		Ley No. 6.196 de 2008 de la Provincia de Chaco	https://perma.cc/23FR-LXSW
		Ley No. 649 de 18 de noviembre de 2004 de la Provincia de Tierra del Fuego, Adhesión de la Provincia a la Ley Nacional No. 25.392, sobre Registro Nacional de Donantes de Células Progenitoras Hematopoyéticas	https://perma.cc/Q589-CR8E
		Ley No. 712 de 29 de noviembre de 2001 de la Ciudad Autónoma de Buenos Aires, de Garantías del Patrimonio Genético Humano	https://perma.cc/EFY9-M3LH

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		Ley No. 8.953 de 19 de septiembre de 2001 de la Provincia de Cordoba, Inviolabilidad del Genoma Humano – Establecimiento y Garantización en la Provincia--Prohibiciones y Sanciones	https://perma.cc/RGD9-KAJ9
	Presidential Decree	Decreto No. 200 de 8 de marzo de 1997, Prohíbense los Experimentos de Clonación Relacionados con Seres Humanos	https://perma.cc/4PSR-JCYC
	Regulatory Decree	Decreto Reglamentario No. 267, sobre el Registro Nacional de Donantes de Células creados por Ley No. 25.392 de 13 de febrero de 2003	https://perma.cc/JF8Q-342Q
Brazil	Constitution	Constituição Federal	https://perma.cc/D8KL-2TYJ
	Statute	Lei No. 9,434, de 4 de Fevereiro de 1997	https://perma.cc/WP43-QU96
		Lei No. 10.973, de 2 de Dezembro de 2004, as amended by Lei No. 13.243, de 11 de Janeiro de 2016	https://perma.cc/3JRM-669G
		Lei No. 11.105 de 24 de Março de 2005	https://perma.cc/3C32-R4YL
	Presidential Decree	Decreto No. 5.591, de 22 de Novembro de 2005	https://perma.cc/66LM-PWJQ
	Regulation	ANVISA, Resolução No. 9, de 14 de Março de 2011	https://perma.cc/3D4Z-QA4W
		ANVISA, Resolução RDC No. 23, de 27 de Maio de 2011	https://perma.cc/879S-96VG
		ANVISA, Resolução RDC No. 29, de 12 de Maio de 2008	https://perma.cc/37WS-LXVX
		ANVISA, Resolução RDC No. 33, de 17 de Fevereiro de 2006	https://perma.cc/4CTR-EKJ9
		ANVISA, Resolução RDC No. 214, de 7 de Fevereiro de 2018	https://perma.cc/J5T2-KEBX
		ANVISA, Resolução RDC No. 508, de 27 de Maio de 2021	https://perma.cc/3WM2-4ZM2
		ANVISA, Resolução RDC No. 771, de 26 de Dezembro de 2022	https://perma.cc/7VKE-WPBW
		CNS, Resolução No. 400, de 17 de Abril de 2008	https://perma.cc/4JVF-XFKL
		CNS, Resolução No. 466, de 12 de Dezembro de 2012	https://perma.cc/4C33-55VS

Regulation of Stem Cell Research: Table of Primary Sources

		Ministério da Saúde, Portaria No. 2.526, de 21 de Dezembro de 2005	https://perma.cc/2BU7-6EKF
	Bill of Law	Senado Federal, Projeto de Lei No. 5.153, de 2020	https://perma.cc/NQ7F-FBGW
Chile	Statute	DFL No. 725 de 1968, Código Sanitario	https://perma.cc/V4QB-MB27
		Ley No. 20.120 de 2006, Sobre la Investigación Científica en el Ser Humano, su Genoma, y Prohíbe la Clonación Humana	https://perma.cc/W76D-JVHE
		Ley No. 20.584 de 2012, Regula los Derechos y Deberes que Tienen las Personas en Relación a su Atención en Salud	https://perma.cc/WL99-RTW2
	Executive Order	Norma Técnica No. 57, Regulación de la Ejecución de Ensayos Clínicos que Utilizan Productos Farmacéuticos en Seres Humanos (Ministerio de Salud, 2001)	https://perma.cc/48QS-Z3P2
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